

High-Frequency Price Signals, Thermal Flexibility, and Renewable Integration: Evidence from China's Electricity Market Reform

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March 28, 2026

Abstract

We examine how the transition from administrative dispatch to a high-frequency electricity spot market reshapes generation, renewable integration, and system operations, using the full-scale market reform in a northern Chinese province as a quasi-natural experiment. Combining a Regression Discontinuity in Differences (RD-DD) design with dynamic structural estimation, we first estimate the reform's reduced-form effects and then use the structural model to interpret the underlying unit-level mechanisms. High-frequency price signals induce real-time “intraday net-load reshaping”: thermal generation falls by 12.1%, while wind and solar accommodation rise by 38.5% and 43.8%, with the largest responses concentrated among older, larger baseload units. Structural recovery of marginal costs and startup costs further suggests that the province's “dual-track” contracting architecture is broadly consistent with competitive spot-market outcomes. Counterfactual simulations imply that the full reform bundle expands renewable generation by 7.77 TWh, reduces thermal generation by 7.69 TWh, and lowers total thermal engineering cost—defined as the sum of fuel and startup costs—by 6.59 billion RMB, of which 4.72 billion RMB is attributable to merit-order dispatch and commitment optimization alone. At the same time, the reform raises consumer payment under market-based pricing, underscoring that engineering-cost savings, market-accounting incidence, and social welfare need not move together—a distinction that is directly relevant to electricity market transitions in developing economies.

Keywords: Electricity Market, Spot Market, RD-DD, Dynamic Structural Estimation, Renewable Integration, Engineering Cost

JEL Classification: Q41, Q48, L94, C14

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1 Introduction

China's twin carbon goals—peak emissions before 2030 and carbon neutrality before 2060—have made renewable energy integration a central policy priority. Yet the structure of the power sector poses a fundamental obstacle. For decades, generation was allocated through a centrally planned dispatch system in which provincial authorities assigned equal-share annual planned-generation hours to thermal generators and set on-grid tariffs administratively. This arrangement provided revenue certainty for the coal-dominated installed base but offered no high-frequency price signals to reward operational flexibility, discipline high-cost incumbents, or facilitate the accommodation of intermittent wind and solar output. As renewable penetration rose sharply after 2010, these limitations became increasingly visible: Auffhammer et al. (2021) document that national average curtailment rates for wind and solar reached 17% and 10.3%, respectively, in 2016.

The institutional response to this curtailment problem was the landmark 2015 electricity-sector reform, Document No. 9 of the State Council. By instructing regulators to “control the middle and liberalize both ends,” the reform established the legal basis for shifting from administratively allocated generation to market-based dispatch. Implementation then proceeded through a sequenced pilot program. This northern province—China's largest coal-producing region and a strategic inter-provincial power exporter—was selected as an early reform laboratory, and on April 1, 2021, it launched a full-scale, continuously operating spot market that abolished administrative dispatch quotas and replaced them with competitive 15-minute nodal price clearing. Section 2 describes the reform sequence in detail.

A central premise of electricity market liberalization—rooted in the foundational theories of peak-load pricing (Steiner, 1957; Boiteux, 1960; Williamson, 1966)—is that high-frequency price signals are essential for allocative efficiency when generation is indivisible and electricity cannot be economically stored. Decarbonization reinforces this logic while also complicating it. As intermittent, zero-marginal-cost wind and solar expand, the center of gravity of intraday price variation shifts, and thermal generators move from baseload producers toward flexible residual balancers. Because starting a thermal unit involves substantial discrete costs and thermal stress, spot markets must price both short-run fuel costs and longer-run commitment frictions—a challenge the classic framework was not designed to address. How market design handles this operational inflexibility, and whether it can efficiently coordinate the substitution between volatile renewables and rigid fossil generation, therefore remains a central question in energy economics. The issue extends well beyond China: across developing power systems where coal remains necessary for reliability even as renewable capacity expands rapidly, spot-market design may shape whether decarbonization proceeds

through efficient integration or persistent curtailment.

This paper examines whether the provincial spot-market reform resolved that tension and, if so, through which unit-level mechanisms and with what consequences for engineering costs and market incidence. Answering this question requires overcoming two empirical challenges: isolating the causal effect of the reform from confounding seasonal demand shocks, and modeling the dynamic frictions—most notably startup costs and state-dependent commitment constraints—that shape unit behavior under competitive dispatch (Reguant, 2014). We therefore develop an integrated empirical framework that combines reduced-form causal inference with dynamic structural estimation. Using a unit-level daily panel of 133 thermal generators spanning 2020–2022, a Regression Discontinuity in Differences (RD-DD) design exploits the April 1, 2021 market launch as a sharp policy threshold, comparing a 90-day window around the threshold in 2021 with the corresponding window in 2020 to difference out seasonal demand patterns (Grembi et al., 2016). The structural analysis then recovers unit-level marginal costs and startup costs from a dynamic optimal-stopping model, enabling disciplined counterfactual engineering-cost accounting.

This paper makes three contributions to the energy economics literature. First, we provide causal evidence that the spot-market reform reduced thermal generation and improved renewable accommodation: daily thermal generation fell by 12.1%, the probability of continuous baseload operation declined by approximately 43%, and wind and solar accommodation rose by 38.5% and 43.8%, respectively. Intraday analysis reveals pronounced duck-curve reshaping—thermal generation is suppressed during solar-peak midday hours and recovers in the evening—while mechanism tests show that thermal displacement is stronger on days with greater renewable resource availability, consistent with real-time merit-order crowding out. These responses are also highly heterogeneous across units: larger and older baseload units account for most of the output reduction at the extensive margin, whereas smaller and more flexible units shift into peaking roles and increase conditional output. The event-study evidence is consistent with the same interpretation, with flat pre-reform coefficients and post-reform effects that build over time. Second, we connect these aggregate shifts to unit-level mechanisms through structural cost recovery. Vuong model-selection tests of market conduct favor Perfect Competition over Cournot alternatives for 87.3% of units, which is consistent with—though does not by itself establish—the disciplining role of the dual-track forward-contracting architecture. The recovered startup costs also rise sharply with unit size—median startup costs for large units exceed those for small units by a factor of four—providing a structural account of the capacity and age heterogeneity in the reduced-form results. Third, counterfactual simulations imply that the full reform bundle expands renewable generation by 7.77 TWh, reduces thermal generation by 7.69 TWh, and lowers thermal engineering cost by 6.59 billion RMB.

Consumer payment, however, rises because the pre-reform administrative benchmark price was set below the efficient shadow cost of generation. Engineering-cost savings, market-accounting incidence, and welfare should therefore be evaluated separately rather than collapsed into a single metric—a distinction that is directly relevant to electricity market transitions in developing economies.

The remainder of this paper is organized as follows. Section 2 provides the institutional background and describes the architecture of the provincial spot market. Section 3 reviews related literature. Section 4 describes the data. Sections 5 and 6 set out the reduced-form and structural estimation frameworks, respectively. Sections 7 and 8 present the corresponding empirical results. Section 9 concludes.

2 Institutional Background

2.1 The Transition to High-Frequency Spot Pricing

China’s pre-reform power sector operated under a rigid planned-dispatch regime whose three key features — administratively assigned generation hours, fixed benchmark tariffs, and no intraday price signal — are the direct institutional antecedents of the behavior the paper studies. China’s power-sector reform evolved through four stages. The first two stages, spanning the 1980s through the 1990s, focused on expanding generation capacity, separating government from enterprise functions, and establishing administrative oversight, all while preserving a vertically integrated structure centered on state control. The third stage, initiated by the 2002 reform, introduced a structural separation of generation from grid operation and sought to foster competition on the generation side through “generation-grid separation” and benchmark pricing, although actual dispatch and price formation remained largely administrative. The fourth stage began with the 2015 Power Sector Reform, which reoriented policy around “controlling the middle and liberalizing both ends,” promoted direct trading and transmission-and-distribution tariff reform, and created the institutional foundation for a transition from administratively allocated generation toward market-based dispatch.

Despite these earlier reforms, the generation sector for many years continued to operate under a rigid planned-dispatch regime. Provincial authorities allocated equal-share “Annual Planned Generation Hours” to thermal generating units to guarantee basic operational equity, and generators were compensated mainly through fixed benchmark on-grid tariffs. Because generation volumes and prices were largely administratively guaranteed, thermal generators faced limited incentives to operate flexibly or adjust output in response to high-frequency supply and demand conditions. As renewable penetration increased, this system became increasingly ill-suited to balancing the inter-

mittency of wind and solar power, contributing to inefficient dispatch, weak thermal flexibility, and substantial renewable curtailment.

Building on the 2015 reform agenda, regulators launched electricity spot-market pilot programs in 2017 to develop day-ahead and real-time trading, implement security-constrained dispatch, and explore the coordination between spot and medium-to-long-term contracts. This province emerged as one of the most instructive pilot cases in this broader reform sequence because its coal-dominated supply structure, rapid renewable expansion, and strategic role in national energy security made it an especially challenging environment for market design. On April 1, 2021, the province officially abolished administrative dispatch quotas and transitioned to a full-scale, continuously operating centralized spot market. This abrupt regulatory shift replaced administratively set production targets with competitive market clearing based on 15-minute high-frequency nodal pricing. Critically, the April 1, 2021 market launch was implemented on a fixed, publicly announced date without unit-specific grandfathering provisions; the policy boundary is administratively sharp, making it suitable for the regression-discontinuity identification strategy developed in Section 5.

2.2 The “MLT + Spot” Dual-Track Settlement Architecture

A distinctive feature of China’s market design is the mandatory dual-track settlement system, under which thermal units must pre-contract a large share of expected output at a fixed forward price before participating in the spot market. This architecture has two consequences central to the paper’s empirical analysis: it disciplines spot-market bidding by limiting the scope for strategic price-setting, and it creates a marginal incentive structure in which every unit of physical output beyond the contracted volume is exposed to the real-time spot price.

To ensure systemic stability and smooth the transition from the planned economy, regulators mandated that thermal generators cover at least 80% of their expected output through MLT forward contracts, while the corresponding threshold for renewable generators was set at 70%.

In the provincial market design, these MLT contracts function strictly as financial Contracts for Differences (CfDs) rather than physical delivery obligations. The physical dispatch of all generation units is pooled and optimized entirely within the centralized spot market through a “full-volume clearing” mechanism. During financial settlement, the spot market acts as a residual balancing mechanism: generators are settled at the pre-negotiated MLT contract price for their contracted volume, and are only exposed to the volatile real-time spot price for their deviation. This dual-track CfD mechanism thereby secures baseline revenue for generators while preserving full marginal price incentives on every unit of physical output. Section 5 exploits this incentive structure

when testing the competitive-conduct hypothesis against Cournot alternatives.

2.3 Spot Market Structure, Bidding, and Price Rules

A first defining feature of China's electricity spot market is the two-stage market structure. The provincial spot market operates through a tightly coordinated sequence of day-ahead and real-time trading under centralized security-constrained dispatch. In the day-ahead stage, market participants submit generation schedules and bids for the following operating day, and the market operator solves a unit-commitment and dispatch problem that determines baseline hourly commitment and output schedules. This day-ahead process is then followed by a real-time balancing market that clears every 15 minutes to accommodate forecast errors, renewable intermittency, and intraday demand fluctuations. In institutional terms, the day-ahead market provides the primary commitment and scheduling benchmark, while the real-time market continuously re-optimizes dispatch around realized system conditions. This two-stage design is central to our analysis because it is precisely the high-frequency interaction between commitment decisions and intraday redispatch that exposes the operational value of flexibility.

A second defining feature is the highly asymmetric bidding structure across technologies. Renewable generators participate essentially as quantity bidders: wind and solar units submit forecast generation profiles rather than strategic price offers and are dispatched with priority subject to system security constraints. Thermal units, by contrast, serve as the residual balancing technology and therefore face a much richer reporting requirement. They must submit segmented supply curves—typically monotone multi-step offers—together with technical operating parameters such as minimum stable output, ramping capability, startup and shutdown parameters, and other unit-specific constraints relevant for security-constrained unit commitment (SCUC) and security-constrained economic dispatch (SCED). These declared engineering constraints matter directly for market outcomes. They determine not only whether a unit is dispatched on the intensive margin, but also whether it is worth committing at all once startup costs, minimum-load constraints, and expected real-time price movements are taken into account. For this reason, the market's formal requirement that thermal units internalize physical operating frictions provides the institutional foundation for our subsequent structural recovery of dynamic startup costs.

A third defining feature is the explicit price bounds the market imposes on spot prices. Because the dual-track architecture (Section 2.2) routes all physical dispatch through the spot market, these bounds directly shape commitment and dispatch incentives for every generator. The spot market operates within a floor of 0 RMB/MWh and a cap of 1,500 RMB/MWh. The zero price floor is economically important. Because

negative prices are not permitted, thermal generators cannot submit negative bids to stay dispatched during periods of exceptionally low residual demand; instead, they must weigh the losses from producing at near-zero spot prices against the discrete cost of shutting down and the option value of restarting later. Conversely, when scarcity emerges, the capped upside of the spot price rewards flexible units that can ramp quickly or cycle profitably into high-price intervals.

The combination of mandatory pre-contracting (which limits strategic price distortion), asymmetric bidding rules (which make thermal costs recoverable), and explicit price bounds (which define the option-value calculation for commitment) creates a setting in which the structural parameters of interest are empirically tractable within our modeling framework.

This institutional anatomy directly motivates the paper’s empirical strategy. The sharp April 1, 2021 implementation date supports the regression-discontinuity identification in Section 5. The CfD settlement architecture, which exposes every marginal unit of output to real-time nodal prices, validates the competitive-conduct assumption tested in the structural estimation of Section 6. And the mandatory disclosure of unit-level engineering constraints through the bidding process makes startup costs and operating constraints structurally recoverable from observed market behavior.

3 Literature Review

This paper sits at the intersection of three literature strands — electricity market liberalization, renewable integration, and dynamic structural estimation of thermal operations — and contributes by connecting them in a unified evaluation of spot-market reform. The liberalization literature documents efficiency gains from market reform but largely predates high-penetration renewable systems; the renewable-integration literature quantifies the merit-order effect but rarely identifies the role of market design; and the structural literature recovers operating constraints but has seldom been embedded in a policy evaluation of developing-economy electricity markets.

The first strand concerns the literature on electricity market liberalization. A large body of work, based primarily on reforms in the United States and Europe, shows that replacing regulated vertically integrated utilities with market-based dispatch can improve productive and allocative efficiency by strengthening marginal-cost incentives and competitive discipline (Joskow, 2008; Davis and Wolfram, 2012; Craig and Savage, 2013; Chan et al., 2017; Mansur and White, 2012). But most of these reform episodes took place before wind and solar generation reached significant penetration. As a result, the canonical evidence is less informative about settings in which marketization occurs under high renewable penetration and large intraday fluctuations in residual demand.

This gap in the canonical reform literature is particularly salient for China and other developing economies. In these systems, the starting point for reform is not mature wholesale competition but administratively coordinated dispatch, regulated tariffs, and limited operational flexibility, while renewable capacity has expanded rapidly in parallel. The question is therefore not only whether market reform raises efficiency in the standard sense, but also whether it does so in an environment where thermal generators must accommodate large and frequent fluctuations in residual demand created by intermittent renewable output.

A growing body of China-focused work addresses this question directly. Li et al. (2019) argue that market-based trading reforms can improve flexibility and reduce renewable curtailment even in a partly regulated system. Guo et al. (2020) traces the post-2015 institutional evolution of spot-market development, emphasizing the role of pilot programs in improving dispatch efficiency and inter-provincial coordination. Liu et al. (2022), using data from Guangdong’s pilot market, provides the most direct market-design evidence, showing that the welfare gains from spot pricing are sensitive to price caps and floors. Chen et al. (2022) extends this line of work to curtailment outcomes using cross-provincial variation in pilot timing and finds that market-based dispatch reduces renewable curtailment by enabling more granular reallocation of generation. Collectively, this literature shows that market design is a first-order determinant of renewable integration outcomes in China, but it does not trace the unit-level operating mechanisms — commitment flexibility, intraday load-following, and extensive-margin cycling — through which market reform generates those outcomes. Our paper contributes by combining a sharp regression-discontinuity design around the April 2021 implementation date with unit-level evidence on commitment, intraday output profiles, and startup-cost heterogeneity.

The reform efficiency literature establishes the allocative benefits of market pricing, but says little about the specific mechanism — thermal displacement under rising renewable penetration — that is central to this paper; the second strand addresses that mechanism directly. The central mechanism in this literature is the merit-order effect: because wind and solar generation have near-zero variable costs, their entry at the bottom of the merit order pushes higher-cost thermal plants off the supply stack and depresses equilibrium spot prices. A large body of work quantifies this effect across multiple European markets, documenting that renewable generation reduced average wholesale prices by EUR 8/MWh in Germany in 2006 alone, with the effect growing nearly fivefold as capacity expanded (Sensfuß et al., 2008; Würzburg et al., 2013; Cludius et al., 2014). The same intermittency that drives this price suppression also concentrates the thermal fleet’s operational burden in ramp-intensive intraday intervals, raising the value of commitment flexibility as a constraint on how much renewable output can be absorbed.

However, the same physical intermittency that makes renewable generation inexpensive at the unit level also creates system-level integration costs. Joskow (2011) argues that standard levelized-cost comparisons systematically understate these costs by ignoring the economic value of dispatchability: an intermittent unit cannot guarantee output at peak demand and therefore imposes reserve requirements and cycling costs on complementary units — costs that do not appear in the renewable unit’s own accounts. Gowrisankaran et al. (2016) formalize this point in a structural model of the Arizona power market, showing that the social value of additional solar capacity declines substantially once intermittency is priced, because solar output is concentrated in periods of relatively low marginal value and because residual demand becomes more volatile as penetration rises. A related phenomenon is “value cannibalization”: as the installed base of wind and solar expands, these units generate simultaneously with growing frequency, depressing spot prices at precisely the times they produce and eroding the revenues needed to sustain investment. López Prol et al. (2020) document this effect in the California wholesale market, showing that market revenue per unit of renewable output falls monotonically with penetration. It is precisely this thermal flexibility — and whether spot-market reform can unlock it — that the paper’s reduced-form identification in Section 5 targets.

This integration challenge is particularly acute in China, where a coal-heavy fleet designed for baseload operation confronts rapidly growing wind and solar capacity. Davidson et al. (2020) analyze the barriers to wind integration on China’s provincial grid and show that accommodation is constrained primarily by the thermal fleet’s operational inflexibility and the absence of price signals that reward fast-ramping units. Our paper extends this evidence in two ways. First, the RD-DD design — comparing a 90-day window around the April 2021 launch to the same window in 2020 — provides sharper identification than cross-provincial timing variation by directly controlling for seasonal demand confounds. Second, by decomposing the reform effect across the extensive margin (unit commitment), the intensive margin (intraday output profiles), and the structural recovery of startup costs, the paper traces the unit-level primitives — including the differential response of large baseload and small flexible units to spot-price incentives — that aggregate reform effects can obscure.

The reduced-form reform and integration evidence leaves open the question of the unit-level operating primitives behind these aggregate effects; the third strand, on structural estimation of electricity-market behavior, provides the methodological tools this paper draws on. The dominant branch of this literature uses static models of firm behavior — recovering marginal costs from bids, quantities, and prices — to assess the degree to which observed conduct departs from competitive benchmarks. Studies such as Puller (2007) and Hortaçsu and Puller (2008) demonstrate that rich market data identify firms’ marginal costs, strategic markups, and the degree of market

power embedded in observed supply schedules. This literature establishes an empirical framework for measuring conduct in restructured electricity markets, but its one-period perspective is incomplete when firms face important intertemporal operating frictions.

That limitation is especially important in electricity generation, where the production technology is non-convex and current operating decisions depend on future conditions. Startup costs, shutdown decisions, and ramping constraints imply that a generator's choice today cannot be understood solely in terms of contemporaneous prices and quantities. Reguant (2014) shows that startup costs create intertemporal complementarities in bidding and that omitting them biases inference on costs, markups, and market power. Related work by Cullen (2013) likewise emphasizes that policy and price shocks affect generator behavior through dynamic operating margins, because thermal units cannot adjust output costlessly or instantaneously. This dynamic perspective is especially relevant in renewable-heavy systems, where volatility in residual demand raises the value of thermal flexibility and makes the timing of commitment and shutdown decisions central to market outcomes.

Our paper builds on both the static and dynamic literatures. Relative to the static market-power tradition, we move beyond one-period bidding and dispatch decisions to characterize the intertemporal operating margin of thermal units. Relative to the dynamic literature, we embed the structural recovery of startup costs and operating frictions in a broader policy-evaluation framework centered on spot-market reform. By linking reduced-form policy variation to a structural characterization of operating constraints, we show how spot-market reform reshapes unit commitment, thermal flexibility, and renewable accommodation in a developing power system with high renewable penetration—and trace the cost and welfare implications of those changes. A central structural result is that Vuong model comparisons favor Perfect Competition over Cournot alternatives for 87.3% of thermal units, while startup costs scale roughly fourfold between small and large units. These findings connect the institutional design of Section 2 to the unit-level behavioral primitives recovered in Section 6. This linkage between market design, operating constraints, and reform outcomes is, to our knowledge, not established elsewhere for a high-penetration renewable system in a developing economy.

A related body of work examines the distributional and welfare consequences of electricity market liberalization. This literature documents that market-price volatility and tariff design can affect consumer surplus and distributional incidence in ways that diverge from aggregate efficiency measures (Borenstein et al., 2002). Our paper contributes to this strand by documenting that engineering-cost savings and consumer payment can move in opposite directions when the pre-reform administered price is set below the efficient shadow cost of generation — a configuration common in developing-economy power sectors where regulated tariffs chronically underprice

electricity.

Methodologically, this paper applies the regression-discontinuity in differences (RD-DD) framework of Grembi et al. (2016) to evaluate a sharp, calendar-date policy threshold. The credibility of this design requires that units just inside and outside the treatment window are comparable on observable and unobservable dimensions in the absence of the reform, and that no other policy change coincides with the April 1, 2021 implementation date. Section 5 provides supporting evidence on both requirements.

4 Data

The data used in this study come from multiple sources within the provincial power system and together form a unit-level daily panel spanning 2020 to 2022. Unit-level data on generating capacity and output come from the State Grid’s provincial electric power company and record each unit’s county-level location, nameplate capacity, commissioning date, and combustion technology type. We aggregate 15-minute output records for all thermal, wind, and solar units operating within the province to daily generation volumes. To control for exogenous weather shocks affecting both electricity supply and aggregate demand, we incorporate meteorological data from the China Meteorological Administration. We construct county-level weather measures by calculating the geodesic distance from each county centroid to nearby weather stations and retaining all stations within an 80-kilometer radius. If no station falls within that radius, we retain the nearest station. We then remove high-altitude stations whose weather conditions are unlikely to represent local population centers. Because county-level elevation data are unavailable, we identify such stations through within-county comparisons: when a county is matched to multiple stations, we exclude those whose elevation differs from the average elevation of the other matched stations by more than 1,000 meters. Finally, we aggregate station-level observations to the county-hour level using inverse-squared-distance weights,

$$w_{cs} = \frac{1/d_{cs}^2}{\sum_i 1/d_{ci}^2},$$

where w_{cs} denotes the weight assigned to station s for county c , and d_{cs} is the distance between county c and station s . We then aggregate the resulting county-hour observations to the daily level. The final dataset contains county-level hourly and daily measures of temperature, wind speed, solar duration, precipitation, and humidity.

Table 1 reports descriptive statistics for the main variables. Panel A reports cross-sectional characteristics of the thermal units, Panel B summarizes 15-minute market transaction records, and Panels C and D report daily panel statistics for thermal and

renewable units, respectively, split by pre- and post-reform periods.

Panel A summarizes the basic attributes of the 133 thermal units in our sample. Average installed capacity is 345 MW, with a standard deviation of 165.771 MW and a range of 50–660 MW, spanning the full spectrum from small cogeneration units to large baseload generators. Average commissioning age is 14.2 years, indicating that most units are in their mature operational phase.

Panel B presents the 15-minute high-frequency market clearing data that underpin our structural estimation. Average MLT traded volume is 31.820 MWh per 15-minute interval, while average day-ahead spot transaction volume is slightly negative (−0.981 MWh), confirming that generators rely primarily on forward contracts and use the spot market for marginal adjustments. Nodal spot prices average 426.047 RMB/MWh with a standard deviation of 341.092 RMB/MWh, and the maximum observed price reaches the administrative cap of 1,500 RMB/MWh, reflecting substantial intraday price volatility.

Panel C documents changes in the operational behavior of thermal units before and after the reform using daily aggregated data. On the demand side, average daily system load increased from 2,370.126 GWh pre-reform to 2,537.797 GWh post-reform. On the supply side, average daily thermal generation per unit declined slightly, from 3,849.447 MWh to 3,818.643 MWh. The most striking shift appears in the binary operating-status indicator. Prior to the reform, the average operating probability was 0.951, consistent with the baseload-dispatch norm of the planned system. Post-reform, this probability dropped to 0.749 and its standard deviation widened from 0.216 to 0.433, suggesting a shift from near-continuous baseload operation toward demand-driven commitment under spot pricing.

Panel D reports generation for renewable units before and after the reform. Average daily wind generation was essentially flat (381.207 MWh pre-reform vs. 377.189 MWh post-reform), while solar generation rose from 204.373 MWh to 236.354 MWh. Standard deviations for both technologies approach or exceed their respective means, reflecting the inherent intermittency of renewable output. The increase in solar generation partly reflects improved accommodation under market-based dispatch.

Taken together, these descriptive statistics show that despite an upward trend in system load, thermal units shifted away from baseload operation after the reform, with a substantially lower probability of remaining online. Renewable generation improved concurrently. These descriptive patterns are consistent with the prediction that market-based dispatch improves resource allocation, but the causal estimates in the next section are needed to separate reform effects from contemporaneous changes in demand and weather.

Finally, one important pre-processing step warrants mention. Given that high-frequency operational data in power systems may contain extreme outliers driven

by metering faults, extreme meteorological conditions, or abnormal unit testing (as evidenced by the anomalous negative values and excessively large maxima in Table 1), we winsorize the generation variables at the 1st and 99th percentiles prior to conducting the core empirical regressions. This ensures that the estimated treatment effects reflect the average impact of the reform rather than isolated extreme observations.

5 Reduced Form Estimation Framework

We construct an empirical framework that moves from a baseline Regression Discontinuity in Differences (RD-DD) model to event-study analyses, mechanism tests, and heterogeneity assessments.

Given that high-frequency daily and hourly electricity dispatch data are non-negative, right-skewed, and often contain many zeros, especially for renewable output and for units that are offline, traditional log-linear models are not well suited to this setting. As shown by Chen and Roth (2024), applying log-like transformations such as $\log(1 + Y)$ or the inverse hyperbolic sine to outcomes with zeros introduces arbitrary unit dependence, so the resulting semi-elasticities need not represent true percentage changes when treatment affects the extensive margin. We therefore estimate the primary regressions using Poisson Pseudo-Maximum Likelihood (PPML) (Gourieroux et al., 1984) with the fixed-effects structure described below, which yields well-defined semi-elasticities in this setting.

5.1 Baseline Model: Regression Discontinuity in Differences (RD-DD)

To identify the causal effect of the spot market operation, we exploit the official launch of the spot market settlement on April 1, 2021, as a quasi-natural experiment. Evaluating policies where time acts as the running variable poses distinct empirical challenges. As Hausman and Rapson (2018) point out, a standard Regression Discontinuity in Time (RDiT) design is frequently confounded by underlying autoregressive processes, time-series unobservables, and secular trends and seasonal patterns, which can severely bias the estimates.

To isolate the discontinuous policy effect from these temporal confounders, we adopt a Regression Discontinuity in Differences (RD-DD) design. By combining the discontinuity at the policy threshold with a Difference-in-Differences setup (Grembi et al., 2016), we difference out baseline temporal heterogeneity. This quasi-experimental strategy has been used in recent applied microeconomics work to address time-based confounding (Avdic and Karimi, 2018; Persson and Rossin-Slater, 2024).

Table 1: Descriptive Statistics

Variable	Obs.	Mean	Std. Dev.	Min	Max
Panel A: Basic Information of Thermal Units (Cross-Sectional)					
Nameplate Capacity (MW)	133	345.000	165.771	50.000	660.000
Commissioning Age (Years)	133	14.233	5.831	3.000	35.000
Panel B: High-Frequency Market Transactions (15-Minute)					
MLT Traded Volume (MWh)	8,409,117	31.820	24.172	-97.966	718.517
Day-Ahead Traded Volume (MWh)	9,635,949	-0.981	25.218	-662.933	203.307
Nodal Spot Price (RMB/MWh)	9,615,360	426.047	341.092	0.000	1500.000
Thermal Generation (15-Min, MWh)	9,642,816	34.203	52.936	-95200.000	250.000
Total System Generation (15-Min, MWh)	9,642,816	6639.189	997.718	-89900.000	19227.992
Panel C: Thermal Units Daily Panel (Pre- and Post-Reform)					
<i>Pre-Reform</i>					
Generation (MWh)	58,279	3849.447	2568.843	-173.750	16350.595
Operating Status (0/1)	58,279	0.951	0.216	0.000	1.000
Average Temperature (°C)	58,279	8.333	10.736	-24.021	31.247
Average Humidity (%)	58,279	54.078	19.816	6.775	99.745
Precipitation (mm)	58,279	1.251	4.830	0.000	90.854
Average Wind Speed (m/s)	58,279	2.166	0.883	0.119	8.448
Solar Duration (Hours)	58,279	7.522	3.597	0.000	14.093
System Load (GWh)	58,279	2370.126	241.381	1933.833	2950.109
<i>Post-Reform</i>					
Generation (MWh)	84,446	3818.643	3403.744	-84420.834	48400.000
Operating Status (0/1)	84,446	0.749	0.433	0.000	1.000
Average Temperature (°C)	84,446	12.105	10.800	-21.063	33.106
Average Humidity (%)	84,446	57.283	19.418	12.492	99.914
Precipitation (mm)	84,446	1.815	6.686	0.000	181.897
Average Wind Speed (m/s)	84,446	2.176	0.880	0.177	8.917
Solar Duration (Hours)	84,446	6.774	3.928	0.000	14.002
System Load (GWh)	84,446	2537.797	246.528	0.000	3085.422
Panel D: Renewable Units Daily Panel (Pre- and Post-Reform)					
<i>Pre-Reform</i>					
Wind Generation (MWh)	100,537	381.207	333.691	-30.000	4423.872
Solar Generation (MWh)	54,957	204.373	152.819	-2.430	1016.400
<i>Post-Reform</i>					
Wind Generation (MWh)	174,682	377.189	410.549	-2806.938	6158.122
Solar Generation (MWh)	80,762	236.354	198.159	-11.000	1320.000

Notes: This table presents the descriptive statistics for the key variables spanning the 2020-2022 dataset. Panel A illustrates the cross-sectional characteristics of the thermal fleet. Panel B reports the 15-minute high-frequency market transactions and nodal prices. Panels C and D split the daily operational records of thermal and renewable units, respectively, into pre-reform and post-reform periods. The presence of extreme minimum and maximum values (e.g., in generation) necessitates the 1st and 99th percentile winsorization applied prior to the core empirical regressions.

Specifically, analogous to the temporal matching strategy employed by Fu and Gu (2017) to evaluate holiday-specific policies across different years, we define 2021 as the treatment year and use the corresponding period in 2020 as the control group. To ensure exact calendar matching and align weekly cyclicalities, the dates in the 2020 control group are artificially advanced by exactly 364 days, aligning the day-of-week composition between the treatment and control periods.

The baseline PPML RD-DD model is specified as follows:

$$E[Y_{ijt} | X] = \exp(\beta \cdot \text{TreatYear}_j \times \text{PostPeriod}_t + f(\tilde{t}) + \gamma X_{ijt} + \mu_{ij} + \theta_{im} + \tau_{dow} + \lambda_{hol}) \quad (1)$$

where Y_{ijt} denotes the daily power generation of unit i on day t in year j . For solar PV generation, we construct the dependent variable using the total output during 05:00–20:00, corresponding to the approximate sunrise–sunset window around the summer solstice in the province, to exclude zero-output nighttime hours from the dependent variable. To mitigate the influence of extreme outliers, the dependent variable is winsorized at the 1st and 99th percentiles, and negative values are replaced with zero.

The primary variable of interest is the interaction term $\text{TreatYear}_j \times \text{PostPeriod}_t$. Its coefficient, β , captures the net average treatment effect of the market reform. $f(\tilde{t})$ represents the normalized polynomial trends of the running variable (distance in days to April 1). We restrict the estimation sample to a symmetric ± 90 -day window around the policy cutoff. Because a wider bandwidth may capture diverging secular trends across different years, we augment the standard RD polynomials by introducing interactions between the treatment year (TreatYear_j) and linear and quadratic time trends (t, t^2). This specification isolates the discontinuous policy effect from baseline temporal deviations.

The vector X_{ijt} includes the same set of weather and demand controls used in the baseline regressions: average wind speed together with its quadratic and cubic terms, solar duration and its quadratic term, average temperature and its quadratic term, the natural logarithm of actual system load, a heating-season indicator, and daily new confirmed COVID-19 cases. We also directly control for the 2020 trial-market periods to prevent contamination of the comparison year.

The model incorporates unit-by-year (μ_{ij}) and unit-by-month (θ_{im}) fixed effects to control for annual capacity changes and seasonal maintenance schedules. Day-of-week (τ_{dow}) and holiday (λ_{hol}) fixed effects account for recurrent demand fluctuations. Standard errors are clustered at the daily level to allow for arbitrary intra-period correlation of shocks across units.

5.2 Dynamic Effects (Event Study)

To validate the identifying assumptions of the RD-DD model and trace the dynamic adjustment path after the reform, we estimate an event-study specification. This dynamic analysis is useful because market participants may revise operating rules gradually after a major institutional change rather than transition immediately to a new equilibrium (Etgeton et al., 2023). Following the same design as the baseline, we restrict the sample to a symmetric window of 10 weeks before and 10 weeks after the reform date.

Given the non-negative and skewed nature of generation outcomes, we continue to estimate the model by Poisson Pseudo-Maximum Likelihood (PPML) (Gourieroux et al., 1984; Chen and Roth, 2024) with the same fixed-effects structure as in the baseline specification. The event-study specification is

$$E[Y_{ijt} \mid X] = \exp \left(\sum_{k \in K} \beta_k (\text{TreatYear}_j \times \mathbf{1}[\text{RelWeek}_t = k]) + \gamma X_{ijt} + \mu_{ij} + \theta_{im} + \tau_{dow} + \lambda_{hol} \right) \quad (2)$$

Here, Y_{ijt} denotes the generation of unit i on day t in year j , and $\mathbf{1}[\text{RelWeek}_t = k]$ is an indicator for the week relative to the reform date. To preserve power while allowing flexible dynamics near the cutoff, we bin the tails of the event window. Specifically, the left tail pools all weeks with $k \leq -5$, and the right tail pools all weeks with $k \geq 5$. We estimate separate coefficients for $k \in \{-4, -3, -2, 0, 1, 2, 3, 4\}$ and omit $k = -1$ as the reference period.

In the implemented regressions, X_{ijt} includes average wind speed, its quadratic term, its cubic term, the natural logarithm of actual system load, solar duration, the square of solar duration, average temperature, the square of average temperature, a heating-season indicator, a COVID-19 disruption dummy, and flexible time controls given by TreatYear_j , $\text{PostPeriod}_t \times t$, $\text{PostPeriod}_t \times t^2$, $\text{TreatYear}_j \times t$, and $\text{TreatYear}_j \times t^2$. All regressions include the same fixed effects as in the baseline model—unit $\times$ year (μ_{ij}), unit $\times$ month (θ_{im}), day-of-week (τ_{dow}), and holiday (λ_{hol})—with standard errors clustered at the daily level. The coefficients on the pre-reform indicators provide a direct test of parallel trends, while the post-reform coefficients trace the evolution of the treatment effect over time.

5.3 Robustness Check: Placebo Test

To provide additional evidence that the estimated impact of the spot market reform is causal and not driven by unobserved seasonal trends or random shocks, we conduct a placebo test based on randomization inference. If the baseline results in Equation (1) capture the true policy effect, we should not observe significant treatment effects when the reform date is artificially assigned to non-reform periods.

We implement this test through the following procedure:

- **Randomization of Event Dates:** We restrict the potential placebo policy dates to the window between June 1, 2021, and October 1, 2021. This interval is selected to avoid overlap with the actual reform implementation (April 1) while ensuring that the placebo tests operate within the same calendar year to maintain comparable structural conditions.
- **Iterative Estimation:** We randomly draw a pseudo-event date t_{fake} from this interval. For each draw, we construct a corresponding symmetric ± 90 -day window and generate a placebo intervention variable $\text{PostPeriod}_t^{fake}$. We then re-estimate the baseline PPML specification defined in Equation (1), replacing the true interaction term with the placebo interaction $\text{TreatYear}_j \times \text{PostPeriod}_t^{fake}$.
- **Distribution Analysis:** We repeat this process 500 times to generate a distribution of placebo coefficients ($\hat{\beta}^{placebo}$).

The placebo model retains the identical set of controls—including wind speed, temperature, solar duration, system load, and heating indicators—and the same fixed effects (unit-by-year, unit-by-month, day-of-week, and holiday) used in the main analysis.

The placebo test supports identification if two conditions hold. First, the distribution of placebo estimates should center around zero, indicating that the model does not generate spurious effects in the absence of the actual policy. Second, the true benchmark coefficient ($\hat{\beta}^{true}$) estimated using the actual April 1st cutoff should lie well outside the bulk of the placebo distribution. Figure 2 visualizes these results by plotting the histogram and kernel density of the 500 placebo coefficients alongside the true estimate for thermal, wind, and solar generation.

The results broadly support a causal interpretation. For **wind generation** (Panel B), the placebo distribution is approximately symmetric and centered near zero, and the true coefficient ($\hat{\beta}^{true} = 0.326$) falls well into the right tail with an empirical p-value of 0.000—no random placebo cutoff produces an effect as large as the actual reform estimate. For **solar generation** (Panel C), the placebo distribution exhibits a leftward shift relative to zero, likely reflecting the natural seasonal decline in solar radiation that occurs when pseudo-event windows span into the autumn months. Despite this shift, the true coefficient ($\hat{\beta}^{true} = 0.364$) remains clearly beyond the right tail of the distribution, yielding an empirical p-value of 0.024. For **thermal generation** (Panel A), the placebo distribution is tightly concentrated near zero and the true coefficient ($\hat{\beta}^{true} = -0.123$) lies in the left tail, with an empirical p-value of 0.074. While this falls short of the conventional 5% significance threshold, it lies below the 10% threshold,

indicating that a placebo reduction of this magnitude is uncommon under randomly assigned cutoffs.

Taken together, the placebo exercise supports the interpretation that the estimated treatment effects are driven by the timing of the actual reform rather than by seasonal variation or arbitrary cutoff assignment.

5.4 Sensitivity Analysis: Choice of Bandwidth

The estimation of the RD-DD model relies on the selection of a specific time window (bandwidth) around the policy cutoff. In our baseline specification, we use a symmetric window of ± 90 days. To ensure that our main findings are not driven by this specific choice or sensitive to short-run fluctuations, we conduct a bandwidth sensitivity analysis.

We re-estimate the baseline PPML specification (Equation 1) using a range of alternative bandwidths. Specifically, we vary the estimation window from ± 30 days to ± 120 days in 15-day increments (i.e., $h \in \{30, 45, 60, 75, 90, 105, 120\}$). For each bandwidth, we retain the same fixed effects (unit-by-year, unit-by-month, day-of-week, and holiday) and weather and demand controls to maintain consistency with the baseline model.

To assess the bandwidth sensitivity of the design, we plot the treatment coefficients together with their 95% confidence intervals across alternative windows. Stability across bandwidth choices would indicate that the estimated reform effect is not driven by a particular choice of estimation window.

5.5 Mechanisms: Inter-source Substitution and Intraday Flexibility

The core function of the spot market is to reallocate generation through high-frequency price signals. We examine the mechanisms behind the reform's impact along two dimensions: inter-source substitution and intra-thermal operational flexibility.

5.5.1 Inter-source Substitution (Crowding-out Effect)

To identify whether the spot market accommodates renewable energy by crowding out thermal generation, we test whether the reduction in thermal output is larger during periods of greater renewable resource availability.

Variable Centering Strategy: To facilitate the interpretation of interaction terms, we center the renewable-resource variables used in the mechanism regressions. Let c_wind_{ijt} denote wind speed net of its sample mean and let c_ssd_{ijt} denote solar duration net of its sample mean. Centering ensures that the coefficient on the main policy term

can be interpreted at average renewable conditions rather than at the mechanically arbitrary value of zero.

We estimate four regressions in total: two using renewable generation as the dependent variable (wind in Column 1, solar in Column 2) and two using thermal generation (Columns 3 and 4). Each regression includes a single resource interaction, so the wind-interaction and solar-interaction tests are estimated on separate models rather than a single joint specification. The common structural form is:

$$E[Y_{ijt} | X] = \exp \left(\beta_0(\text{TreatYear}_j \times \text{PostPeriod}_t) + \beta_1(\text{TreatYear}_j \times \text{PostPeriod}_t \times z_{ijt}) + \delta z_{ijt} + \gamma X_{ijt} + \mu_{ij} + \theta_{im} + \tau_{dow} + \lambda_{hol} \right), \quad (3)$$

where $z_{ijt} = c\text{-wind}_{ijt}$ in Columns 1 and 3, and $z_{ijt} = c\text{-ssd}_{ijt}$ in Columns 2 and 4. The vector X_{ijt} includes temperature, temperature squared, log system load, heating-season, Covid-19, and trial-period indicators, together with second-order RD polynomial trends in the running variable fully interacted with *PostPeriod* and *TreatYear* to allow flexible local trends on each side of the cutoff. The thermal specifications (Columns 3 and 4) additionally include the other resource variable (solar duration in Column 3, wind speed in Column 4) and a weighted coal-price index as controls, ensuring that the estimated interaction coefficient is not confounded by cross-resource correlation or input-cost variation. As required by a valid triple-interaction specification, all lower-order interaction terms ($\text{PostPeriod}_t \times z_{ijt}$ and $\text{TreatYear}_j \times z_{ijt}$) are included in every specification but suppressed in the reported table. All models are estimated by PPML with standard errors clustered at the daily level.

The coefficient β_0 captures the reform effect on the relevant generation type at mean resource conditions. For renewable-generation outcomes (Columns 1 and 2), the interaction coefficient $\beta_1 > 0$ indicates that the post-reform increase in renewable output is amplified on days with stronger natural-resource conditions. For thermal-generation outcomes (Columns 3 and 4), $\beta_1 < 0$ indicates that the crowding-out of thermal generation is larger when renewable supply is more abundant—consistent with the spot market enabling real-time substitution of thermal generation by renewable sources.

5.5.2 Intraday Flexibility and Ramping

To characterize the physical operational shifts induced by high-frequency price signals, we extend the analysis from daily aggregate generation to hourly operational flexibility. The spot market is expected to shift thermal units from baseload generation to load-following roles, accommodating renewable intermittency.

Ramping Capability. We first examine the intensive margin of flexibility by constructing hourly ramping indicators. Let Y_{ijht} denote the generation of unit i in hour h on day t in year j . We define the hourly ramp rate as $\Delta Y_{ijht} = Y_{ijht} - Y_{ij,h-1,t}$. Based on this, we construct three dependent variables: (1) *Absolute Ramping* ($|\Delta Y_{ijht}|$), measuring the total magnitude of adjustment; (2) *Ramp-up* ($\max\{\Delta Y_{ijht}, 0\}$); and (3) *Ramp-down* ($\max\{-\Delta Y_{ijht}, 0\}$).

Given the non-negative and highly skewed nature of these ramping variables, we estimate the hourly models by PPML (Gourieroux et al., 1984; Chen and Roth, 2024). The specification follows the same RD-DD logic as the baseline model, using a symmetric ± 90 -day window around the reform date. The regressions include the same core control structure adapted to the hourly data, along with unit-by-year (μ_{ij}), unit-by-month (θ_{im}), day-of-week (τ_{dow}), and holiday (λ_{hol}) fixed effects. Standard errors are clustered at the daily level. Formally, for each ramping outcome the estimated equation is

$$E[R_{ijht} | X] = \exp\left(\beta (\text{TreatYear}_j \times \text{PostPeriod}_t) + \gamma X_{ijht} + \mu_{ij} + \theta_{im} + \tau_{dow} + \lambda_{hol}\right), \quad (4)$$

where $R_{ijht} \in \{|\Delta Y_{ijht}|, \max\{\Delta Y_{ijht}, 0\}, \max\{-\Delta Y_{ijht}, 0\}\}$ denotes the relevant ramping outcome for unit i in hour h on day t in year j , and X_{ijht} collects the weather, demand, and second-order RD polynomial-trend controls adapted to the hourly panel.

Cycling Frequency. Beyond continuous output adjustments, the spot market may also affect discrete commitment decisions at the extensive margin. We define a unit as operating in hour h on day t of year j if its hourly generation exceeds 10% of nameplate capacity when capacity information is available; otherwise, we apply a fixed absolute-output threshold. Let D_{ijht} denote this binary operating-status indicator. A startup occurs when $D_{ijht} = 1$ and $D_{ij,h-1,t} = 0$, while a shutdown occurs when $D_{ijht} = 0$ and $D_{ij,h-1,t} = 1$. We aggregate these hourly transitions to the daily level and construct three outcomes: the number of startups, the number of shutdowns, and total daily cycling.

In the data, these daily cycling outcomes are sparse and concentrated at zero. In addition, we cap multiple within-day startups or shutdowns at one event per day, which further compresses the support of the dependent variables. In this setting, PPML frequently fails to converge for cycling regressions. We therefore estimate the effect on cycling using a linear RD-DD model with the same fixed effects and control structure as in the baseline specification:

$$N_{ijt}^k = \beta \cdot \text{TreatYear}_j \times \text{PostPeriod}_t + \gamma X_{ijt} + \mu_{ij} + \theta_{im} + \tau_{dow} + \lambda_{hol} + \epsilon_{ijt} \quad (5)$$

where N_{ijt}^k denotes daily startups, shutdowns, or total cycling for unit i on day t in

year j . Given the zero-dominated support of these outcomes, the linear specification provides a transparent and numerically stable measure of the change in expected daily transition frequency.

Intraday Heterogeneity. Finally, to characterize how the reform reshapes the intraday profile of generation, we move from daily outcomes to hourly generation and estimate a separate regression for each hour of the day. Specifically, for each $h \in \{0, \dots, 23\}$, we re-estimate the hourly generation equation on the subsample with $\text{Hour} = h$, using hourly generation as the dependent variable and maintaining the same RD-DD logic, controls, fixed effects, and daily clustering structure. The resulting sequence of hour-specific treatment coefficients traces when thermal, wind, and solar generation respond most strongly over the day. Rather than estimating one pooled regression with hour interactions, we run 24 separate hour-specific regressions and visualize the intraday redistribution of generation under spot-market pricing.

5.6 Heterogeneity: Capacity, Age, and Margins

Faced with identical market price signals, thermal units exhibit heterogeneous responses determined by their physical constraints and cost structures. To uncover the micro-foundations of the aggregate policy effect, we decompose the treatment effect along both operational margins and unit characteristics.

5.6.1 Margins of Adjustment

We first decompose the total reduction in thermal generation into the *extensive margin* (the decision to operate) and the *intensive margin* (the level of output conditional on operating). This decomposition clarifies whether the market reform primarily drives the complete shutdown of marginal units or reduces the output of units that remain active.

Let $D_{ijt} = \mathbb{1}(Y_{ijt} > 0)$ denote the binary operational status of unit i on day t in year j . We estimate the policy effect on three outcomes using the PPML estimator with the full set of fixed effects $(\mu_{ij}, \theta_{im}, \tau_{dow}, \lambda_{hol})$:

- **Total Effect:** The dependent variable is unconditional daily generation Y_{ijt} , including zeros.
- **Extensive Margin:** The dependent variable is the binary indicator D_{ijt} . Estimated by PPML, this specification captures the change in the probability that a unit is dispatched.
- **Intensive Margin:** The dependent variable is Y_{ijt} estimated on the subsample with $Y_{ijt} > 0$, capturing how generation changes conditional on operation.

5.6.2 Heterogeneity by Unit Characteristics

To examine how the policy effect varies across unit characteristics, we conduct split-sample analyses based on two key physical attributes: installed capacity and commissioning age.

Installed Capacity (Economies of Scale). Larger units generally possess higher thermal efficiency and lower marginal costs. We stratify the sample into three discrete groups based on nameplate capacity: *Small* units with capacity below 300 MW, *Medium* units with capacity between 300 MW (inclusive) and 600 MW (exclusive), and *Large* units with capacity at least 600 MW. We estimate the baseline RD-DD model separately for each subsample to test whether smaller, less efficient units bear a disproportionate share of the market-induced output reduction.

Unit Age (Commissioning Age). Newer units are typically equipped with more advanced technology, which may improve flexibility and reduce emission rates. We measure age as years since commissioning. To capture the vintage effect, we employ a median split: units at or above the sample median age are classified as *Old*, while those below are classified as *Young*. By comparing the coefficients across these subsamples, we examine whether the spot market shifts generation away from older units and toward newer fleets.

For both capacity and age heterogeneity, we report the Total, Extensive, and Intensive margin effects within each subgroup to fully characterize the adjustment dynamics.

6 Structural Estimation Framework

While the reduced-form Regression Discontinuity in Differences (RD-DD) model identifies the aggregate causal impacts of the spot market reform, it does not recover the underlying microeconomic primitives—marginal costs, startup costs, and commitment frictions—that drive generator behavior. To decompose these effects and evaluate the engineering-cost consequences of alternative dispatch rules, we estimate a structural model of generator competition. The estimation proceeds in four sequential steps: (1) estimating residual demand elasticity, (2) recovering unobserved marginal costs and testing market conduct, (3) approximating dynamic startup costs using a B-spline framework, and (4) simulating counterfactual dispatch scenarios.

6.1 Step 1: Nonparametric Estimation of Residual Demand Elasticity

This step constructs a unit-level measure of residual demand elasticity, which is the key input for recovering unit-level marginal costs and testing market conduct in the subsequent analysis. Our identification strategy is grounded in the New Empirical Industrial Organization (NEIO) framework. As formally established by Bresnahan (1982) and Lau (1982), the degree of market competitiveness and unobserved marginal costs can be econometrically identified from observed price and quantity data, provided there are exogenous demand rotations. This approach has been widely applied to test market power since the early work of Porter (1983). Following Wolak (2003) and Puller (2007), we adopt a flexible nonparametric approach based on Gaussian kernel methods to accommodate nonlinearities in the 15-minute high-frequency bidding data.

Identification and construction. Our identification strategy uses the standard short-run approximation that aggregate demand is locally price-inelastic, $D'(p) \approx 0$. Under this approximation, the slope of the residual demand curve facing unit i is determined entirely by the supply responses of rival units. Following the empirical strategy developed by Reguant (2014) for wholesale electricity markets, rather than estimating residual demand directly, we recover it from underlying supply slopes. This approach improves numerical stability and clarifies the mapping from unit-level behavior to market outcomes. In particular, the residual demand derivative is defined as:

$$D'_{r,i}(p_t) = - \sum_{j \neq i} S'_j(p_t) = - (S'_{market}(p_t) - S'_i(p_t)), \quad (6)$$

where $S'_{market}(p_t)$ denotes the slope of the aggregate supply curve at the market-clearing price p_t .

Estimation algorithm. We implement Equation (6) in three stages. First, for each unit i , we construct the expected supply curve $\hat{S}_i(p_t)$ using a Nadaraya–Watson kernel estimator based on the unit's historical operating observations $(p_h, q_{i,h})$ matched by calendar month, hour-of-day, day-of-week, and heating season. With a Gaussian kernel $K(\cdot)$ and bandwidth w , the base supply estimator evaluated at price p_t is defined as:

$$\hat{S}_i(p_t) = \frac{\sum_h K_{i,t} q_{i,h}}{\sum_h K_{i,t}}, \quad (7)$$

where $K_{i,t} = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{p_t - p_h}{w}\right)^2\right)$.

The supply slope $\hat{S}'_i(p_t)$ is then obtained by taking the analytical derivative of this kernel estimator with respect to price p_t . Applying the quotient rule, the slope estimator

is:

$$\hat{S}'_i(p_t) = \frac{\left(\sum_h K'_{i,t} q_{i,h}\right) \left(\sum_h K_{i,t}\right) - \left(\sum_h K_{i,t} q_{i,h}\right) \left(\sum_h K'_{i,t}\right)}{\left(\sum_h K_{i,t}\right)^2}. \quad (8)$$

The bandwidth w^* is selected via out-of-sample cross-validation over the discrete candidate set $W \in \{10, 30, 50, 80, 100, 150, 200, 300, 400\}$ (RMB/MWh), minimizing the mean squared prediction error of generation quantities. To capture intraday and seasonal non-stationarities, we run the CV search independently for each scenario (defined by the combination of calendar month, hour-of-day, day-of-week, and heating season indicator) whenever at least 200 observations are available; otherwise, the globally cross-validated bandwidth—selected as 80 RMB/MWh on the pooled data—serves as a fallback. This bandwidth is then used for both the level estimator $\hat{S}_i(p_t)$ and its derivative $\hat{S}'_i(p_t)$. Next, at each time t , we aggregate unit-level slopes to obtain the market stiffness, $S'_{\text{market}}(p_t) = \sum_i \hat{S}'_i(p_t)$. Finally, we compute $D'_{r,i}(p_t)$ by netting out unit i 's own slope from $S'_{\text{market}}(p_t)$ as in Equation (6) and construct the residual demand elasticity as $\epsilon_{i,t} = D'_{r,i}(p_t) \times (p_t/q_{i,t})$. To limit the influence of numerical noise in sparse price regions, computed elasticity values are winsorized at ± 1000 , a threshold substantially wider than the economically plausible range for a 162-unit market.¹ Standard errors for $\hat{\epsilon}_{i,t}$ are computed via a date-block bootstrap: calendar date blocks are resampled with replacement across $B = 50$ replications; the kernel slopes $\hat{S}'_i$ are fixed at their point estimates and only the market-aggregation step $\sum_j \hat{S}'_j$ is re-executed.^{2 3}

6.2 Step 2: Marginal Cost Recovery and Market Conduct

6.2.1 Firm Optimization and First-Order Conditions

With the nonparametrically estimated elasticities, we recover the unobserved physical marginal costs (MC) using a structural model of profit maximization. Generators

¹The three sample sizes used across the paper reflect different selection criteria. The reduced-form analysis uses a balanced panel of 133 thermal units—those with continuous daily records in both the pre- and post-reform windows. The structural estimation universe comprises 162 thermal units that appear in the 15-minute spot-market transaction data, including some units excluded from the balanced reduced-form panel due to missing daily coverage. After applying the sequential testing-down GMM procedure, 114 of the 162 units converge to stable cost estimates and are used in the counterfactual simulations; the remaining 48 units exhibited insufficient price variation or failed convergence checks and are excluded from the structural analysis.

²This two-step bootstrap approximation captures aggregation-level sampling uncertainty conditional on the estimated slopes. It does not propagate uncertainty from bandwidth selection or the kernel estimation itself. This is standard practice when the first-step nonparametric estimator is computationally intensive.

³All kernel computations are GPU-accelerated via CUDA to handle the 9.6-million-observation sample. As a robustness check, we also estimate residual demand elasticity via a Panel IV (2SLS) approach, instrumenting price with temperature and competitor system load; results are consistent with the nonparametric benchmark.

maximize their profits across both the spot market and medium-to-long-term (MLT) forward markets. Generator i maximizes static profit with a predetermined forward contract position Q_{it}^c executed at contract price P_{it}^c :

$$\max_{q_{it}} \Pi_{it} = p_t(Q) \cdot (q_{it} - Q_{it}^c) + P_{it}^c Q_{it}^c - C_i(q_{it}) \quad (9)$$

Taking the first-order condition (FOC) with respect to q_{it} yields the generalized pricing rule adjusted for contract coverage:

$$p_t = MC_{it}(q_{it}; \theta) + \frac{1}{|\epsilon_{it}|} \frac{q_{it} - Q_{it}^c}{q_{it}} p_t \quad (10)$$

To limit numerical instability during model inversion when residual demand elasticity approaches zero or when firms hold deep short positions, we impose structural boundaries guided by the physical and regulatory limits of the provincial spot market (which enforces a 0 to 1,500 RMB/MWh price collar). Specifically, the contract exposure ratio $(q_{it} - Q_{it}^c)/q_{it}$ is bounded within $[-2.0, 1.0]$, and the overall physical exposure markup term is bounded within $[-0.9, 4.0]$. We retain the full-year sample in estimation; combined heat and power (CHP) units face binding heat-load constraints in winter that shift their supply behavior, but these constraints are partly absorbed by the coal-price scaling factor ϕ_{it} together with the full-year temperature and wind instruments, which remain valid demand-side shifters across seasons.

To account for substantial fuel price volatility during the sample period, the physical marginal cost is specified as a polynomial function scaled by a daily standardized coal price factor ϕ_{it} :

$$MC_{it} = \phi_{it} \left(\alpha_1 + \alpha_2 q_{it} + \alpha_3 q_{it}^2 \right) \quad (11)$$

where $\phi_{it} = P_{it}^{coal} / \bar{P}_i^{coal}$, with P_{it}^{coal} denoting the unit-day coal input price and $\bar{P}_i^{coal}$ representing the unit's sample mean. This separable specification effectively isolates daily fuel-price shocks from the baseline physical technology parameters governed by $\theta = (\alpha_1, \alpha_2, \alpha_3)$.

6.2.2 GMM Estimation Framework

We estimate the structural parameter vector θ using the Generalized Method of Moments (GMM). Let y_{it} denote the structurally implied, fuel-adjusted marginal cost under the assumed behavioral model. Under Perfect Competition, $y_{it} = p_t / \phi_{it}$; under Unit Cournot, $y_{it} = [p_t - \text{Markup}_{it}] / \phi_{it}$, where $\text{Markup}_{it} \equiv p_t(q_{it} - Q_{it}^c) / (q_{it} |\epsilon_{it}|)$ is the conduct-implied markup. The structural residual is defined as:

$$\varepsilon_{it}(\theta) = y_{it} - \left(\alpha_1 + \alpha_2 q_{it} + \alpha_3 q_{it}^2 \right) \quad (12)$$

To address the endogeneity of real-time generation q_{it} , we construct an instrumental variable vector $\mathbf{Z}_{it} = (1, \tilde{T}_{it}, \tilde{W}_{it})$, where $\tilde{T}_{it}$ denotes standardized local temperature and $\tilde{W}_{it}$ denotes standardized local wind speed. Temperature is a valid demand-side instrument year-round: it shifts both summer cooling load and winter heating load, generating exogenous variation in the market-clearing price without directly affecting a unit's physical cost curve. Wind speed is a supply-side instrument: it shifts wind power generation and hence system net load and thermal clearing prices, without altering individual thermal units' marginal cost schedules. Together, these two instruments provide independent sources of exogenous price variation across all seasons. The population moment conditions are $E[\mathbf{Z}_{it}\varepsilon_{it}(\boldsymbol{\theta})] = \mathbf{0}$. The GMM estimator minimizes the quadratic objective function:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \left(\frac{1}{N} \sum_{i,t} \mathbf{Z}_{it} \varepsilon_{it}(\boldsymbol{\theta}) \right)' \mathbf{W} \left(\frac{1}{N} \sum_{i,t} \mathbf{Z}_{it} \varepsilon_{it}(\boldsymbol{\theta}) \right) \quad (13)$$

where $\mathbf{W}$ is the weighting matrix, initialized as the identity matrix $\mathbf{I}$ in the first step. We employ an iterated GMM procedure (up to five iterations until convergence) where $\mathbf{W}$ is replaced at each step by the inverse of a Heteroskedasticity and Autocorrelation Consistent (HAC) covariance matrix $\hat{\mathbf{S}}$, estimated using a Newey–West kernel with lag length $\lfloor 4(T/100)^{0.25} \rfloor$. Furthermore, to compute robust standard errors for hypothesis testing, we implement a block-bootstrap procedure at the daily level to account for within-day serial correlation.

6.2.3 Sequential Testing-Down Algorithm

To avoid overfitting baseload units with noisy generation variation, we implement a sequential “testing down” procedure based on iterative t-tests for structural model selection. For each unit, the estimation algorithm proceeds as follows:

1. **Quadratic Specification:** We first estimate the full model $\boldsymbol{\theta} = (\alpha_1, \alpha_2, \alpha_3)$. If the t-statistic for the quadratic term α_3 meets the 95% confidence threshold ($|t| > 1.96$), the quadratic functional form is retained.
2. **Linear Specification:** If α_3 is statistically insignificant, the model automatically down-selects to a linear specification, estimating only (α_1, α_2) . If the slope parameter α_2 is significant ($|t| > 1.96$), the linear model is selected.
3. **Constant Specification:** If both α_3 and α_2 fail to reject the null hypothesis, the model falls back to a constant marginal cost specification (α_1), reflecting a completely flat marginal cost curve.

6.2.4 Market Conduct Identification

To identify the form of market conduct, we recover MC curves under three distinct, non-nested behavioral assumptions: (1) **Perfect Competition (PC)** ($P_t = MC_{it}$); (2) **Unit Cournot-Nash**; and (3) **Firm Collusive Cournot**, where generators internalize price effects on the parent firm's total portfolio, substituting $(q_{it} - Q_{it}^c)$ with $(Q_{ft} - Q_{ft}^c)$, in which $Q_{ft} = \sum_{j \in f} q_{jt}$ is the firm's total generation and $Q_{ft}^c = \sum_{j \in f} Q_{jt}^c$ is the firm's total forward contract position.

Because these three alternative specifications of market conduct represent non-nested structural models, standard nested likelihood ratio tests are conceptually invalid. We therefore compare these competing models using the Vuong (1989) test based on the Kullback–Leibler Information Criterion (KLIC). Because our marginal cost parameters are estimated via GMM rather than Maximum Likelihood, we extend this testing logic using the generalized non-nested model selection framework of Rivers and Vuong (2002), which adapts the Vuong test to models defined by moment conditions.

Operationally, we implement this non-nested test independently for each thermal unit using its 15-minute high-frequency operational data. For any two competing behavioral models, M_1 (e.g., Perfect Competition) and M_2 (e.g., Unit Cournot), we compute the difference in their squared structural residuals at each time period t , denoted as $d_t = \hat{\varepsilon}_{2,t}^2 - \hat{\varepsilon}_{1,t}^2$. The unit-level test statistic is then constructed as:

$$Z = \sqrt{n} \frac{\bar{d}}{s_d} \xrightarrow{d} N(0, 1) \quad (14)$$

where n is the number of valid joint intraday observations for the unit, $\bar{d}$ is the sample mean of the loss differences, and s_d is the sample standard deviation of d_t .

We apply a two-tailed decision rule to evaluate relative fit. If $Z > 1.96$, Model 1 yields a significantly better fit; if $Z < -1.96$, Model 2 fits better; and intermediate values ($-1.96 \leq Z \leq 1.96$) indicate that the two competing models are statistically indistinguishable. We aggregate these unit-level test outcomes across the thermal fleet. If the Perfect Competition model fits better than the Cournot alternatives for most units, we interpret this pattern as consistent with limited strategic markups under extensive forward-contract coverage. This comparison provides the basis for adopting the competitive assumption ($P = MC$) in the subsequent structural estimation of dynamic startup costs.

However, interpreting these recovered marginal costs solely through static FOCs has important limitations in wholesale electricity markets. As shown by Reguant (2014), the static assumption ignores key intertemporal frictions—most notably discrete startup costs. If a generator rationally bids below its static marginal cost during low-demand periods to avoid the discrete penalty of a shutdown-restart cycle, a static

framework would misclassify this as apparent “negative markups.” Thus, while the static competitive assumption is useful for recovering the baseline intensive-margin cost curve, fully characterizing allocative efficiency requires embedding these recovered costs in a dynamic optimal-stopping framework. This consideration motivates our structural recovery of dynamic startup costs in Step 3.

6.3 Step 3: Dynamic State Transitions and Startup Cost Estimation

6.3.1 Dynamic Optimization Problem

The spot market heavily penalizes inflexibility, making discrete startup costs a critical determinant of generator behavior. We formulate the operational decision as a dynamic optimal stopping problem, following the dynamic discrete-choice framework of Rust (1987). The generator maximizes the expected discounted value by choosing the operating status $\sigma_{it} \in \{0, 1\}$:

$$V(S_t, \sigma_{it-1}) = \max_{\sigma_{it} \in \{0,1\}} \begin{cases} \pi_{it}(S_t) - SC_i \cdot \max(0, \sigma_{it} - \sigma_{it-1}) + \delta E[V(S_{t+1}, 1) | S_t] & \text{if } \sigma_{it} = 1 \\ \delta E[V(S_{t+1}, 0) | S_t] & \text{if } \sigma_{it} = 0 \end{cases} \quad (15)$$

where SC_i represents the unobserved startup cost, and $\delta = (1.06)^{-1/35,040} \approx 0.9999983$ is the per-period discount factor corresponding to a 6% annual discount rate applied over 15-minute intervals, with $35,040 = 4 \times 24 \times 365$ denoting the number of 15-minute intervals per calendar year.⁴

The flow payoff $\pi_{it}(S_t)$ is the realized static profit evaluated at the optimal generation level $q_{it}^*(P_t)$. Using the marginal cost parameters $\theta = (\alpha_1, \alpha_2, \alpha_3)$ recovered in Step 2 (adjusted by the daily coal price ratio), the optimal static generation is obtained by solving $P_t = MC_i(q_{it})$ for q_{it} , subject to physical operating constraints. Specifically, $q_{it}^*(P_t)$ is constrained to lie in the interval $[0.25 \cdot Cap_i, Cap_i]$, where $0.25 \cdot Cap_i$ represents the minimum stable operating load and Cap_i represents nameplate capacity. The static profit is therefore calculated as the spot market revenue net of the integral of the marginal cost curve (Total Variable Cost):

$$\pi_{it}(S_t) = P_t \cdot q_{it}^*(P_t) - \int_0^{q_{it}^*(P_t)} MC_i(x; \theta) dx - \bar{f} \cdot Cap_i \cdot \Delta t \quad (16)$$

where $\bar{f} = 5$ RMB/MW-hour is a capacity-related fixed operating cost and $\Delta t = 0.25$ hours is the period length.

The state vector $S_t = (P_t, L_t)$ evolves according to two separate stochastic processes. Aggregate thermal generation L_t follows an AR(1) process driven by exogenous weather

⁴The 6% annual rate corresponds to the typical weighted average cost of capital (WACC) for Chinese thermal power enterprises.

conditions $\mathbf{W}_t$.⁵

$$L_t = \alpha_L + \rho_L L_{t-1} + \gamma_L \mathbf{W}_t + \mu_m + \tau_h + \epsilon_{L,t} \quad (17)$$

where μ_m and τ_h are month and hour-of-day fixed effects. Because electricity spot prices exhibit pronounced regime behavior—switching between low-price baseload intervals and high-price peak intervals— P_t is modeled as a hybrid Markov-switching process:

$$P_t = (1 - \rho_P) \mu_{s_t} + \rho_P P_{t-1} + \tau_h + \epsilon_{P,t}, \quad \epsilon_{P,t} \sim N\left(0, (1 - \rho_P)^2 \sigma_{s_t}^2\right) \quad (18)$$

where $s_t \in \{1, 2\}$ is a latent regime governed by a first-order Markov chain, μ_{s_t} and $\sigma_{s_t}^2$ are regime-specific mean and variance estimated by maximum likelihood, $\rho_P \approx 0.3$ is an AR(1) persistence parameter, and τ_h is an hour-of-day fixed effect. An AR(1) price specification serves as a robustness check and as a computational fallback for units for which the Markov-switching estimation fails to converge.

6.3.2 B-spline Approximation and Hybrid Estimation Algorithm

Dynamically simulating continuous state spaces over an $H = 192$ -period (48-hour) horizon suffers from the curse of dimensionality. To resolve this computational bottleneck, we employ the Conditional Choice Probability (CCP) approach pioneered by Hotz and Miller (1993). Instead of solving the full dynamic programming problem via backward induction, we approximate the forward-looking value functions using a flexible B-spline probit framework (Ryan, 2012). The nonlinear relationship among potential profits, system load, and the commitment decision is captured through quantile-based B-spline basis functions $B_k(\cdot)$:

$$\Pr(\sigma_{it} = 1) \approx \Phi \left(\sum_{k=1}^K \gamma_k B_k(\pi_{it}^{\text{pot}}) + \sum_{k=1}^K \lambda_k B_k(L_t) + \psi \sigma_{it-1} + \boldsymbol{\eta} \mathbf{W}_t + \mu_i + \tau_h \right) \quad (19)$$

where π_{it}^{pot} denotes the potential operating profit if the unit were online at period t (evaluated at the simulated price using the recovered marginal cost parameters), μ_i is a unit fixed effect, and τ_h is an hour-of-day fixed effect. To ensure convergence in the presence of sparse state transitions for certain baseload units, we implement an adaptive dimensionality reduction strategy: the algorithm initializes with a third-degree polynomial using five quantile knots, and reduces to a second-degree or first-degree specification if matrix singularity is detected.

Forward value simulation ($H = 192$). Using the estimated state-transition models—AR(1) for thermal load and Markov Regime Switching for price—we compute the

⁵In estimation, L_t is divided by 10,000 for numerical stability.

critical threshold governing the startup decision by forward-simulating the continuation values over a 48-hour horizon ($H = 192$). For each unit-period, we simulate $N = 500$ independent paths of future states $\{S_\tau\}_{\tau=t}^{t+H}$. To ensure economic realism and avoid explosive value divergence, the simulated forward price paths are mechanically clipped within the bounds of $[0, 1500]$ RMB/MWh, mirroring the physical price collars of the provincial spot market. The expected discounted value gain from starting relative to waiting is computed as:

$$\Delta V_{it} = E \left[\sum_{\tau=t}^{t+H} \delta^{\tau-t} (\pi(S_\tau, \text{On}) - \pi(S_\tau, \text{Wait})) \right] \quad (20)$$

Identification and SMD estimation. Following Magnac and Thesmar (2002), the model implies a threshold-crossing rule: $\text{Start}_{it} = 1 \iff \Delta V_{it} > SC_i + \eta_{it}$, where η_{it} is an idiosyncratic preference shock distributed as $N(0, \sigma^2)$. We recover the latent startup cost SC_i by minimizing the distance between the reduced-form B-spline probability $\hat{\text{Pr}}_{\text{spline}}(\text{Start}_{it} = 1 \mid S_t)$ and the structural probability $\Phi((\Delta V_{it} - SC_i)/\sigma)$. This yields a Simulated Minimum Distance (SMD) estimator implemented via Weighted Non-Linear Least Squares (WNLLS):

$$\widehat{SC}_i = \arg \min_{SC_i} \sum_t \omega_{it} \left(\hat{\text{Pr}}_{\text{spline}}(\text{Start}_{it} = 1 \mid S_t) - \Phi \left(\frac{\Delta V_{it} - SC_i}{\sigma} \right) \right)^2 \quad (21)$$

To achieve the asymptotically optimal minimum distance estimator, the unit-time specific weights are constructed as the inverse variance of the choice probabilities: $\omega_{it} = \varphi(Z_{it})^2 / [\text{Pr}_{it}(1 - \text{Pr}_{it})]$, where $\varphi(\cdot)$ denotes the standard normal PDF, $Z_{it} \equiv (\Delta V_{it} - SC_i)/\sigma$ is the probit index, and Pr_{it} is the predicted choice probability. This weighting scheme downweights observations with high variance in the reduced-form CCPs, anchoring identification on observations where the reduced-form choice probability is precisely estimated.

To accommodate heterogeneous cycling frequencies across the fleet while maintaining numerical stability, we employ a two-branch estimation strategy:

- **Branch A (Individual Estimation):** For flexible units with sufficient observed transition histories ($N_{\text{trans}} \geq 30$), we directly estimate the unit-specific scalar SC_i and the idiosyncratic shock variance σ_i .
- **Branch B (Pooled Estimation):** For rarely-run, ultra-rigid units ($N_{\text{trans}} < 30$), we impose a parsimonious physical prior by projecting SC_i onto observable technological attributes, effectively pooling the SMD moments across capacity

tiers. The pooled specification is estimated as:

$$SC_{\text{pooled},i} = \beta_0 + \beta_{\text{cap}} \left(\frac{\text{Capacity}_i}{100} \right) + \beta_{\text{age}} \left(\frac{\text{Age}_i}{10} \right) \quad (22)$$

Finally, we form a unified, unit-level startup-cost series by merging $\widehat{SC}_i^A$ from Branch A and $\widehat{SC}_{\text{pooled},i}$ from Branch B. For hypothesis testing and constructing confidence intervals, the standard errors of the recovered startup costs are computed from the asymptotic covariance matrix, numerically approximated via the inverse of the Hessian of the WNLLS objective function evaluated at the optimum, scaled by the mean squared error of the structural residuals.

6.4 Step 4: Counterfactual Simulation Algorithms

The structural parameters recovered in Steps 1–3 enable us to quantify the engineering-cost and renewable-integration consequences of alternative dispatch and commitment rules. Throughout, the simulations abstract from exogenous physical contingencies—planned maintenance, forced outages, and detailed transmission network constraints—and should be interpreted as disciplined engineering-accounting counterfactuals rather than literal operating forecasts. We organize the counterfactual exercise around four analytically distinct margins: (i) the *intensive margin*, which captures efficiency gains from within-fleet merit-order re-dispatch holding the set of online units fixed; (ii) the *extensive margin*, which captures the additional gain from endogenous unit commitment—shutting down expensive units that the spot market’s startup-cost signals make uneconomical to run; (iii) the *renewable-integration margin with fixed commitment*, which quantifies how much additional renewable energy can be absorbed through within-commitment redispatch when renewable availability expands; and (iv) the *renewable-integration vs. fully optimized benchmark*, which measures the joint gain from co-optimizing commitment and dispatch under a higher-renewable operating environment. The five scenarios described below map directly onto these four margins.

We implement a unified counterfactual pipeline that combines reduced-form renewable shocks with structural thermal-system reoptimization. Rather than directly imposing a counterfactual thermal-load series, we first reconstruct total system demand at 15-minute frequency, then construct counterfactual *renewable availability*, and finally endogenize thermal output, renewable utilization, and curtailment under alternative operational constraints. This design makes the decomposition economically transparent: differences across scenarios arise from changes in dispatch rules, commitment rules, or renewable availability, while the underlying system demand is held fixed.

Step 4.1: Constructing counterfactual renewable availability. The simulation starts from reduced-form estimates of the reform-induced change in renewable generation. Let Y_{kht}^{Obs} denote observed hourly generation of renewable technology $k \in \{\text{Wind}, \text{Solar}\}$ in hour h and day t , and let $\hat{\beta}_{kh}$ denote the estimated reform effect. We convert these estimates into an hourly renewable increment, ΔRE_{kht} , using the estimated coefficients together with the observed renewable baseline. For wind, the algorithm first validates whether hourly coefficients are sufficiently consistent with the corresponding daily estimate; if the validation ratio is below a preset threshold, we use the hourly coefficient path, and otherwise revert to a daily-coefficient allocation. For solar, we apply daily coefficient estimates together with a daylight mask that restricts positive solar increments to hours with observed solar irradiance. Hence, the renewable shock is technology-specific and may be estimated at different temporal resolutions across wind and solar.

Let W_{ht}^{Obs} and S_{ht}^{Obs} denote observed hourly wind and solar generation. We construct counterfactual hourly renewable *availability* as

$$W_{ht}^{CF} = \max \left\{ 0, W_{ht}^{Obs} + \Delta RE_{Wind,ht} \right\}, \quad (23)$$

$$S_{ht}^{CF} = \max \left\{ 0, S_{ht}^{Obs} + \Delta RE_{Solar,ht} \right\}, \quad (24)$$

and then expand both series from hourly to 15-minute frequency while preserving each hour's total energy. Counterfactual renewable availability at 15-minute frequency is therefore

$$RE_t^{CF} = W_t^{CF} + S_t^{CF}. \quad (25)$$

Total system demand is reconstructed as observed thermal generation plus observed renewable generation over the modeled horizon:

$$D_t = G_t^{Obs, Thermal} + W_t^{Obs} + S_t^{Obs}. \quad (26)$$

Thus, the simulation holds gross demand fixed, allowing the reform to operate through expanded renewable supply that the thermal system must accommodate. In this implementation, the thermal requirement is endogenous and depends on how much of the available renewable energy can actually be absorbed given commitment and minimum-load constraints.

Step 4.2: Fixed-commitment counterfactual accounting. Given a committed thermal fleet, we compute for each 15-minute period t the thermal generation that would be needed if all available renewable energy were absorbed:

$$TH_t^{need} = \max \{ 0, D_t - RE_t^{CF} \}. \quad (27)$$

However, actual thermal generation may differ from TH_t^{need} because committed units are subject to both upper-capacity and minimum-load constraints. Specifically, if i is committed, its feasible dispatch lies in

$$q_{it} \in [\underline{q}_i, \bar{q}_i], \quad \underline{q}_i = 0.25 Cap_i, \quad \bar{q}_i = Cap_i, \quad (28)$$

where the lower bound reflects a uniform minimum-load ratio of 0.25.

For a given commitment set, we first construct the feasible aggregate thermal target,

$$TH_t^{target} = \min \left\{ \sum_i \bar{q}_i, \max \left(TH_t^{need}, \sum_i \underline{q}_i \right) \right\}. \quad (29)$$

Thermal dispatch is then determined either by economic redispatch or by a bounded projection toward observed output, depending on the scenario. Once simulated thermal output $TH_t = \sum_i q_{it}$ is obtained, renewable utilization is computed residually as

$$RE_t^{used} = \min \{ RE_t^{CF}, \max(0, D_t - TH_t) \}. \quad (30)$$

Curtailment follows as

$$Curt_t = RE_t^{CF} - RE_t^{used} \geq 0. \quad (31)$$

Utilized renewable generation is then allocated between wind and solar in proportion to their relative availability in each interval. This accounting distinction matters because the reform does not mechanically imply one-for-one thermal displacement at every period: if committed thermal minimum load is too high, part of the additional renewable availability is curtailed.

Step 4.3: Dispatch and commitment solvers. The simulation uses two structural routines.

First, the economic-dispatch solver implements a lambda-iteration algorithm. For each time interval, the solver dispatches unit-level supply from estimated marginal-cost curves and iteratively updates the system shadow price until the absolute gap between aggregate thermal supply and net thermal demand falls below 0.25 MWh. The solution is clipped at unit-specific lower and upper bounds, and a final adjustment step reallocates any remaining shortage within residual capacity. The implied system clearing price is recovered as the equilibrium shadow price at convergence of the lambda-iteration algorithm (the λ^* at which aggregate thermal supply equals net thermal demand), and then aggregated into a load-weighted average price across dispatch intervals.

Second, the commitment solver for endogenous scenarios operates at the *daily* level. Units are ranked by a unit-level cost metric, and the algorithm selects the lowest-cost

subset of units satisfying two conditions: (i) cumulative committed capacity must cover the day's peak thermal requirement plus a reserve margin, and (ii) cumulative minimum thermal output should not exceed the day's minimum thermal need whenever feasible. The reserve margin is not imposed as an arbitrary constant in the preferred simulations. Instead, we construct a day-specific historical margin map,

$$m_d = \frac{\text{CommittedCap}_d - \text{PeakLoad}_d}{\text{PeakLoad}_d}, \quad (32)$$

truncated from below at 5%, and use this matched historical series in the baseline S2 and counterfactual S5 simulations. Startup costs are also scenario-specific. In S5, startup costs are applied only for units newly brought online relative to the previous day, whereas in the fixed-commitment counterfactual scenarios S3 and S4 the baseline observed startup-cost total is carried over.

Step 4.4: Scenario construction. We implement five scenarios that are directly used in the decomposition.

- **Baseline (Observed Operations).** Thermal generation, renewable generation, and startup behavior are taken from the observed pre-reform data. Gross demand is reconstructed as observed thermal plus observed wind and solar generation over the modeled horizon.
- **S1 (Economic dispatch on observed load with observed commitment).** The commitment set is fixed at the observed running units, while thermal dispatch is re-optimized by the merit-order solver against observed thermal load. This scenario isolates *intensive-margin* gains from better within-fleet dispatch, holding the set of online units fixed.
- **S2 (Endogenous commitment and dispatch on observed load).** The algorithm selects the daily commitment set endogenously using the rank-and-select heuristic with matched historical reserve margins, and then applies economic dispatch. Comparing S1 and S2 identifies *extensive-margin* gains from shutting down relatively inefficient units and reconfiguring the online fleet.
- **S3 (Fixed commitment with reference dispatch under counterfactual renewable availability).** The set of running units is fixed at the observed pre-reform commitment pattern. Thermal dispatch is not solved by merit order; instead, output is projected onto the feasible interval $[q_i, \bar{q}_i]$ while staying as close as possible to observed unit-level dispatch. Renewable utilization and curtailment are then determined endogenously from the residual demand left after this bounded

reference dispatch. Crucially, this is *not* a simple proportional scaling rule: the projection respects each unit's physical bounds and deviates from observed dispatch only to the minimum extent required to accommodate the revised renewable supply.

- **S4 (Fixed commitment with economic redispatch under counterfactual renewable availability).** The commitment set remains fixed at the observed pattern, but thermal output is re-optimized through the merit-order solver subject to the same minimum-load and capacity constraints. Renewable absorption, curtailment, and any residual imbalance are then computed endogenously. The difference between S3 and S4 captures the value of within-commitment redispatch when renewable availability increases.
- **S5 (Endogenous commitment and redispatch under counterfactual renewable availability).** Both commitment and dispatch are re-optimized under the counterfactual renewable-availability path. The commitment stage uses the daily thermal need implied by $D_t - RE_t^{CF}$ together with the matched historical reserve-margin map and startup-cost penalties for newly committed units. This constitutes the fully optimized counterfactual scenario.

Step 4.5: Decomposition logic. The five scenarios map directly onto four analytically distinct margins of cost saving. The first is the *intensive engineering-cost gain*, defined as $Cost_{Baseline} - Cost_{S1}$. By comparing observed operations with a scenario in which thermal output is re-optimized through economic dispatch while holding the committed fleet fixed, this difference isolates the efficiency gain attributable to better within-fleet merit-order allocation, absent any change in the set of online units. The second is the *extensive engineering-cost gain*, defined as $Cost_{S1} - Cost_{S2}$. Moving from fixed-commitment to endogenous commitment identifies the additional saving from shutting down relatively costly units and reconfiguring the online fleet at the daily horizon—the commitment margin that the spot market activates through startup-cost incentives.

The third comparison shifts the focus from the cost of observed operations to the ability of the thermal fleet to absorb additional renewable energy. Contrasting S3 and S4 under counterfactual renewable availability provides a *renewable-integration test with fixed commitment*: holding the pre-reform commitment pattern constant while varying whether thermal output is re-optimized through merit order quantifies how much of the additional renewable supply can be absorbed through within-commitment redispatch, and how much is curtailed because inflexible minimum-load constraints prevent deeper thermal reduction. The fourth and most composite comparison, $Cost_{S2} - Cost_{S5}$, constitutes the *renewable-integration comparison relative to the fully optimized benchmark*. It measures how engineering cost and renewable absorption change when both com-

mitment and dispatch are jointly re-optimized under the reform-induced expansion in renewable availability, incorporating the interaction between a reshaped thermal fleet, an augmented renewable supply path, and endogenous curtailment.

This final comparison is inherently multidimensional: the gap between S2 and S5 reflects the joint effects of thermal displacement, renewable absorption, endogenous curtailment, and the co-optimization of dispatch and commitment under a higher-renewable operating environment. Accordingly, the counterfactual exercise is best interpreted as a structural simulation of how the spot-market reform changes renewable integration and engineering cost through the interaction of renewable availability, curtailment, and thermal operating constraints.

7 Reduced Form Estimation Results

7.1 Baseline Results: The Impact of Spot Market Reform

This section presents the baseline estimates of the reform’s impact on generation. Using the RD-DD strategy, we estimate the model over a symmetric ± 90 -day window around the official market launch date of April 1, 2021. To align day-of-week effects and seasonal cycles, the dates in the 2020 control group are advanced by exactly 364 days. Periods of earlier spot market trials in 2020 are excluded from the sample to prevent baseline contamination.

Table 2 reports the baseline estimates using the Poisson Pseudo-Maximum Likelihood (PPML) estimator. Across specifications, we control for weather and demand conditions, including wind speed and its higher-order terms, together with unit-by-year, unit-by-month, day-of-week, and holiday fixed effects. To assess the sensitivity of the RD-DD design to the specification of the running variable, Columns (1) through (3) report results using alternative polynomial trends in distance to the cutoff. Column (1) includes a linear trend interacted with the post-period indicator, Column (2) adds a quadratic polynomial, and Column (3) uses a cubic specification.

Because the dependent variable is daily generation in levels and the estimator is PPML, the coefficients on $\text{TreatYear}_j \times \text{PostPeriod}_t$ are semi-elasticities. Panel A focuses on thermal generation. Across all three polynomial specifications, the policy coefficient is negative and statistically significant at the 1% level. In Column (3), the coefficient is -0.129 . This implies an average reduction of approximately 12.1% ($\exp(-0.129) - 1 \approx -0.121$) in daily thermal generation after the spot market reform.

Conversely, Panels B and C show that the reform significantly increased renewable generation. Panel B reports results for wind power, with a coefficient of 0.326 (significant at the 1% level) in Column (3), implying a 38.5% ($\exp(0.326) - 1 \approx 0.385$) increase in wind generation attributable to the reform. Panel C shows a positive and significant

impact on solar power. The coefficient of 0.363 (significant at the 1% level) in Column (3) implies a 43.8% ($\exp(0.363) - 1 \approx 0.438$) increase in solar output.

Overall, these baseline results are stable across trend specifications and suggest that the spot market reallocated generation away from thermal units and toward renewables.

Table 2: Impacts of Market Reform (90-Day Window)

Dependent Variable: Daily Generation	(1)	(2)	(3)
Panel A: Thermal Power			
RD-DD Policy Effect	-0.123*** (0.028)	-0.127*** (0.027)	-0.129*** (0.027)
Observations	45,652	45,652	45,652
Pseudo R^2	0.727	0.728	0.728
Panel B: Wind Power			
RD-DD Policy Effect	0.326*** (0.099)	0.321*** (0.097)	0.326*** (0.098)
Observations	81,764	81,764	81,764
Pseudo R^2	0.760	0.762	0.763
Panel C: Solar Power			
RD-DD Policy Effect	0.364*** (0.065)	0.364*** (0.065)	0.363*** (0.064)
Observations	43,396	43,396	43,396
Pseudo R^2	0.859	0.859	0.859
Trend Specification	Linear	Quadratic	Cubic
Controls	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes

Notes: The table reports baseline PPML estimates using a symmetric ± 90 -day window around the April 1, 2021 policy implementation date. Columns (1), (2), and (3) employ linear, quadratic, and cubic polynomial trends of the running variable, respectively. The model aligns dates based on a 364-day shift to ensure exact day-of-week matching. Trial periods are directly excluded. Baseline controls include wind speed and its higher-order terms, temperature, system load, heating-season and COVID indicators, and solar duration controls where applicable. Standard errors, clustered at the daily level, are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

7.2 Dynamic Effects (Event Study)

While the baseline RD-DD estimates capture the average treatment effect of the spot market reform, they do not reveal how responses evolve over time. To examine dynamic adjustment and assess the parallel-trends assumption, we estimate an event-study model. We restrict the sample to a ± 10 -week window around the reform and use binned tails: T_Pre_Long collects all weeks at ≤ -5 and T_Post_Long collects all weeks at ≥ 5 , while individual weekly coefficients are estimated for weeks -4 through 4 . The

week immediately preceding the reform ($t = -1$) is the omitted category. The event-study models are estimated by PPML with the same control structure as in the baseline specification, together with unit-by-year (μ_{ij}), unit-by-month (θ_{im}), day-of-week (τ_{dow}), and holiday (λ_{hol}) fixed effects. Standard errors are clustered at the daily level.

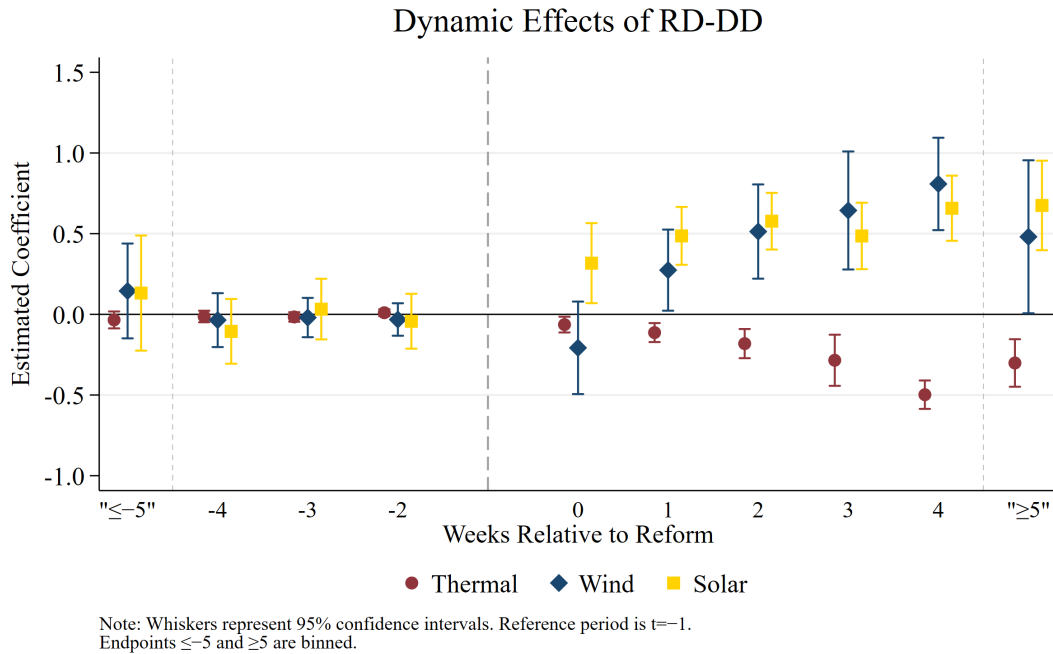


Figure 1: Dynamic Effects of the Spot Market Reform

Notes: This figure plots the estimated coefficients of the dynamic policy effects for thermal, wind, and solar power generation, alongside their 95% confidence intervals. The x-axis represents the number of weeks relative to the reform implementation date. The reference period is the week immediately preceding the market reform ($t = -1$). The endpoints $x = -5$ and $x = 5$ correspond to the binned terms T_Pre_Long (≤ -5 weeks) and T_Post_Long (≥ 5 weeks), respectively; dashed vertical lines at ± 4.5 visually separate the binned endpoints from the individual weekly estimates.

Figure 1 and Table 3 present the event-study results. The pre-treatment coefficients are consistent with the parallel-trends assumption. For all three energy types, every pre-reform term—including the binned T_Pre_Long and the individual weekly terms T_Pre_4 through T_Pre_2 —is statistically indistinguishable from zero at conventional significance levels. The joint Wald test across all pre-treatment terms yields p -values of 0.204 (thermal), 0.410 (wind), and 0.136 (solar), indicating no systematic divergence between the treatment and comparison years prior to the reform. Taken together, the individual coefficients and joint tests support the identifying assumption that, absent the reform, outcomes would have evolved similarly across the two years around the threshold.

Following the implementation of the spot market reform ($t \geq 0$), generation patterns diverge sharply and the effects grow over time. For thermal power (Column 1), the estimated reduction grows in magnitude from -0.113^{***} in week 1 to -0.498^{***} by

week 4, with the binned post-period term T_Post_Long equal to -0.301^{***} . This pattern is consistent with a sustained reduction in thermal output as renewables account for a larger share of dispatch under spot pricing.

Columns (2) and (3) reveal a symmetric and rapid positive response for renewable energy. Wind power generation rises from 0.274^{**} in week 1 to 0.808^{***} by week 4, and the T_Post_Long term remains significant at 0.481^{**} . Solar power shows a similarly strong dynamic, with an immediate positive impact at the time of the reform ($T_Current = 0.317^{**}$) growing to 0.658^{***} in week 4 and sustaining at 0.675^{***} thereafter.

Taken together, these dynamic estimates indicate that the reform-induced shift in the generation mix persisted over the post-reform window. The gradual strengthening of the coefficients is consistent with a progressive reallocation of dispatch toward renewables under spot-market price signals.

Table 3: RD-DD Dynamic Effects

Dependent Variable: Daily Generation Unit Type:	(1) Thermal	(2) Wind	(3) Solar
$T_Pre_Long (\leq -5 \text{ weeks})$	-0.035 (0.027)	0.145 (0.150)	0.132 (0.182)
T_Pre_4	-0.013 (0.018)	-0.036 (0.085)	-0.106 (0.102)
T_Pre_3	-0.016 (0.015)	-0.020 (0.062)	0.032 (0.096)
T_Pre_2	0.010 (0.011)	-0.032 (0.051)	-0.043 (0.087)
$T_Current$ (Week 0)	-0.064** (0.025)	-0.207 (0.146)	0.317** (0.127)
T_Post_1	-0.113*** (0.030)	0.274** (0.128)	0.486*** (0.091)
T_Post_2	-0.181*** (0.046)	0.513*** (0.149)	0.577*** (0.090)
T_Post_3	-0.285*** (0.081)	0.644*** (0.187)	0.486*** (0.105)
T_Post_4	-0.498*** (0.045)	0.808*** (0.146)	0.658*** (0.103)
$T_Post_Long (\geq 5 \text{ weeks})$	-0.301*** (0.075)	0.481** (0.242)	0.675*** (0.142)
Observations	37,153	66,449	35,401
Controls	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes

Notes: The table reports the PPML event-study estimates. The dependent variable is daily generation. The reference period is the week immediately preceding the market reform ($t = -1$). Standard errors, clustered at the daily level, are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

7.3 Robustness Check: Placebo Test

To assess whether the baseline RD-DD estimates reflect the reform rather than unrelated seasonal patterns or random shocks, we conduct a placebo test based on randomization inference. If the reform is driving the observed shifts in the generation mix, placebo treatment dates assigned to non-reform periods should not produce similarly large effects.

Following our empirical strategy, we randomly draw 500 pseudo-event dates from the window between June 1, 2021, and October 1, 2021. For each pseudo cutoff, we construct a symmetric ± 90 -day window, generate a placebo treatment variable, and re-estimate the baseline model with the same controls and fixed effects as in the main specification. This procedure yields an empirical distribution of placebo coefficients ($\hat{\beta}^{placebo}$), which we compare with the benchmark estimate from the actual reform date ($\hat{\beta}^{true}$).

Figure 2 plots the randomization-inference results for thermal, wind, and solar power. The gray bars show the histogram of the 500 placebo estimates, the solid navy line shows the corresponding kernel density, and the dashed maroon line marks the estimate obtained using the actual April 1 cutoff.

Three patterns stand out. First, for wind and thermal generation, the placebo distributions are centered near zero, suggesting that the design does not mechanically generate large treatment effects in periods without the reform. For solar generation (Panel C), the placebo distribution exhibits a leftward shift, likely reflecting the natural seasonal decline in solar radiation when pseudo-event windows extend into the autumn months.

Second, the true estimates lie in the tails of the placebo distributions. For thermal power (Panel A), the benchmark estimate of -0.123 (Column 1, linear-trend specification; the headline cubic-trend estimate is -0.129) falls in the left tail of the placebo distribution, with an empirical two-sided p -value of 0.074. While this does not reach the conventional 5% threshold under a two-sided criterion, the estimate is clearly significant under the one-sided alternative implied by the paper’s directional hypothesis that the reform reduces thermal generation, confirming that placebo effects of this magnitude are uncommon under randomly assigned dates.

Third, the renewable estimates are clearly distinct from the placebo distributions. In Panel B (Wind Power) and Panel C (Solar Power), the true estimates of 0.326 and 0.364 lie in the right tails, with empirical p -values of 0.000 for wind and 0.024 for solar.

Taken together, the placebo exercise supports the interpretation that the main estimates are linked to the timing of the reform rather than to generic seasonal variation or random cutoff assignment. For thermal output, the randomization-inference evidence is somewhat weaker under a two-sided test than for wind or solar generation, but it is

clearly supportive under the directionally relevant one-sided alternative, so the placebo results remain consistent with the paper’s interpretation of a reform-induced decline in thermal generation.

7.4 Sensitivity Analysis: Choice of Bandwidth

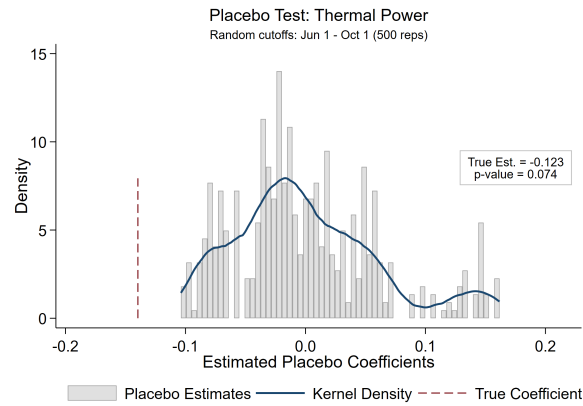
A concern with RD-DD designs is their sensitivity to the choice of estimation window. Wider bandwidths incorporate more observations and increase statistical precision, but risk absorbing pre-existing trends that the polynomial controls do not fully capture. Narrower bandwidths sharpen local identification around the cutoff at the cost of statistical power. The ± 90 -day baseline reflects a judgment about this trade-off; the analysis below examines whether the main findings hinge on that choice.

Figure 3 traces the estimated policy coefficients and their 95% confidence intervals as the symmetric bandwidth varies from ± 30 to ± 120 days in 15-day increments, holding the quadratic polynomial controls, fixed-effect structure, and daily clustering fixed throughout.

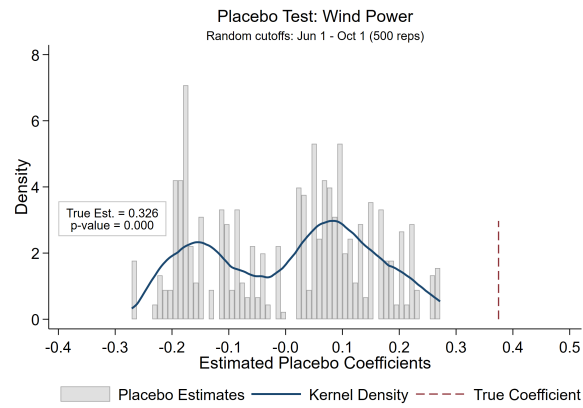
The thermal estimates (top panel) are strikingly stable. Across all seven bandwidths, the coefficient remains negative and statistically significant, ranging from approximately -0.13 to -0.16 . There is no systematic upward or downward drift as the window expands—a pattern that would have emerged if a pre-existing downward trend in thermal generation were being progressively absorbed into the treatment estimate. The baseline coefficient of -0.129 at ± 90 days falls near the center of this range, providing no indication that it is an outlier driven by the particular window chosen for the main analysis.

The renewable estimates are equally stable. For both wind power (middle panel) and solar power (bottom panel), the policy coefficients remain positive across all bandwidth choices. The narrowest window (± 30 days) yields the widest confidence intervals, reflecting the sharp reduction in effective sample size close to the cutoff, but the point estimates are directionally consistent with the baseline. Precision improves monotonically as the window expands, and no sign reversal or large coefficient swing appears at any point in the range. This pattern is inconsistent with the hypothesis that baseline renewable results are artifacts of a particular window choice: if the estimated increase in renewable generation were driven by a temporary local shock rather than a structural change at the reform date, the estimate would erode as the window widened to include calmer surrounding periods.

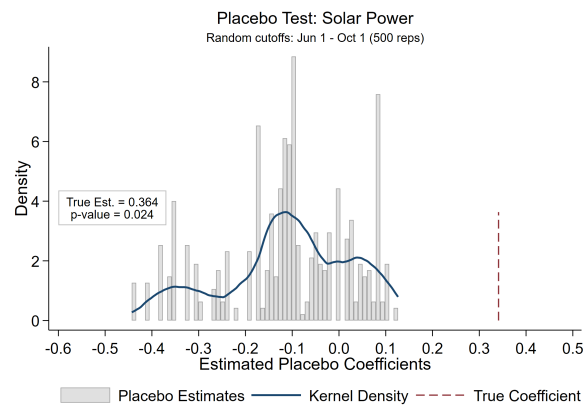
Taken together, the bandwidth sensitivity analysis supports the conclusion that the reform effects on all three generation types are stable features of the data. The direction, approximate magnitude, and statistical significance of the estimated coefficients persist across a wide range of estimation windows, which is consistent with—though does not



Panel A: Thermal Power



Panel B: Wind Power



Panel C: Solar Power

Figure 2: Placebo Tests: Distribution of Estimated Coefficients

Notes: This figure plots the empirical distributions (histogram and kernel density) of the estimated policy coefficients from 500 placebo tests. The pseudo-event dates are randomly assigned between June 1, 2021, and October 1, 2021. The dashed maroon line represents the true estimate derived from the actual reform date. The empirical p -value represents the proportion of placebo estimates whose absolute value exceeds the absolute value of the true estimate.

by itself prove—the identifying assumption that the reform date constitutes a credible discontinuity in the generation mix.

7.5 Mechanisms: Inter-source Substitution and Intraday Flexibility

The baseline results show that the spot market reform shifted the generation mix. This section examines two mechanisms behind that shift: substitution across energy sources and changes in thermal operational flexibility.

7.5.1 Inter-source Substitution (Crowding-out Effect)

If the spot market facilitates renewable accommodation by displacing thermal generation, the strength of this substitution should vary with renewable resource availability. In particular, the reduction in thermal output should be larger when wind or solar conditions are stronger.

To test this mechanism, we estimate four separate augmented RD-DD specifications as described in Section 5.5.1, each including one centered resource variable and its three-way interaction with the policy indicator. The main RD-DD coefficient captures the average treatment effect at mean resource conditions, while the interaction term captures how the policy effect varies with resource availability. The thermal specifications additionally control for the other resource variable and for the coal-price index.

Table 4 reports the results. Columns (1) and (2) examine how the reform changes the sensitivity of renewable generation to resource conditions; Columns (3) and (4) test whether the corresponding crowding-out of thermal generation is stronger when renewable supply is more abundant.

Column (1) shows that the main RD-DD coefficient for wind generation at mean wind-speed conditions is 0.100, but it is not statistically significant. The interaction term, however, is positive and highly significant (0.029, $p < 0.01$), indicating that the policy-induced increase in wind generation is strongly amplified on windier days. This pattern suggests that the benefits of spot pricing for wind accommodation are concentrated in high-resource periods: at typical wind conditions, the reform’s direct effect on wind output is imprecisely estimated, but when wind supply is abundant, the market mechanism channels a substantially larger share of that potential into realized generation.

Column (2) presents a complementary picture for solar power. Unlike wind, solar exhibits a large and precisely estimated main effect: at mean solar-duration conditions, the reform increases solar generation by approximately 47.6% ($\beta_0 = 0.389$; $\exp(0.389) - 1 \approx 0.476$, $p < 0.01$). The interaction term is also positive and significant (0.009, $p < 0.01$), indicating that this baseline gain is amplified on days with greater sunlight exposure. Taken together, Columns (1) and (2) establish that the market reform

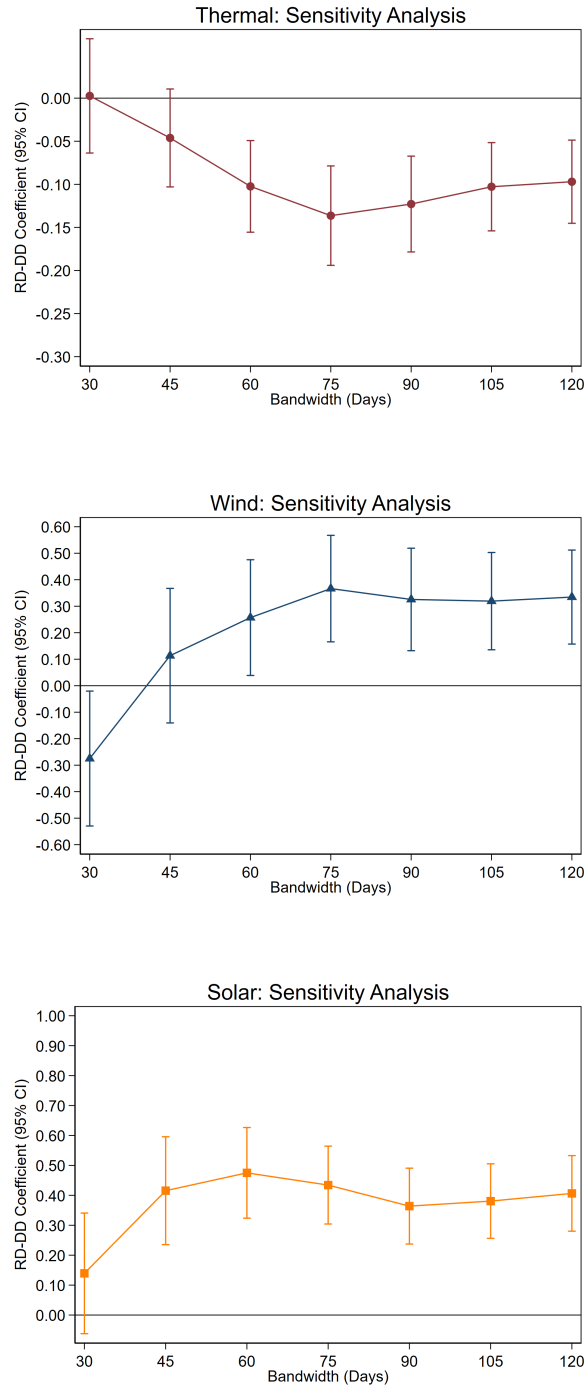


Figure 3: Sensitivity Analysis: Alternative Bandwidths

Notes: This figure plots the estimated RD-DD policy coefficients and their 95% confidence intervals across varying symmetric bandwidths ranging from ± 30 to ± 120 days. The models are estimated using PPML with weather and demand controls and unit-by-year, unit-by-month, day-of-week, and holiday fixed effects. Standard errors are clustered at the daily level. The horizontal solid black line represents zero.

improves the ability of the system to translate renewable resource availability into realized generation, with the mechanism operating robustly across both wind and solar technologies.

Columns (3) and (4) document the corresponding displacement of thermal generation. In Column (3), the interaction between the policy indicator and centered wind speed is -0.012 ($p < 0.01$), implying that thermal output falls by more on windier days after the reform—precisely the days on which wind generation rises most sharply. Column (4) shows the same mechanism for solar: the interaction with solar duration is -0.002 ($p < 0.01$), so thermal generation is reduced by more on sunnier days. The main RD-DD coefficients at mean resource conditions are -0.101 and -0.162 in Columns (3) and (4), respectively, both significant at the 1% level, confirming that baseline thermal displacement is present even on average-resource days.

Across all four columns, the pattern is internally consistent: the reform amplifies the sensitivity of the generation mix to renewable resource conditions, expanding output from wind and solar units when natural resources are abundant while simultaneously displacing thermal generation. This reflects the real-time substitution mechanism that competitive spot markets are designed to enable.

Table 4: Mechanism Analysis - Resource Allocation Efficiency (90-Day Window)

Dependent Variable: Generation Type: Mechanism:	(1) Wind Sensitivity	(2) Solar Sensitivity	(3) Thermal (Wind Crowd-Out)	(4) Thermal (Solar Crowd-Out)
RD-DD Policy Effect (at mean)	0.100 (0.081)	0.389*** (0.039)	-0.101*** (0.023)	-0.162*** (0.033)
Wind Speed (Centered)	-0.001 (0.001)		-0.000 (0.000)	
Policy $\times$ Wind Speed	0.029*** (0.003)		-0.012*** (0.001)	
Solar Duration (Centered)		0.001 (0.000)		-0.000* (0.000)
Policy $\times$ Solar Duration		0.009*** (0.001)		-0.002*** (0.000)
Observations	81,764	43,396	45,652	45,652
Pseudo R^2	0.806	0.887	0.731	0.729
Controls	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes

Notes: Natural resource variables (wind speed, solar duration) are mean-centered. The main effect represents the policy impact under average weather conditions; the interaction term represents how that effect varies with resource availability. Standard errors, clustered at the daily level, are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

7.5.2 Intraday Flexibility and Ramping

To dissect the physical operational shifts induced by the market—following the empirical strategy described in Section 5.5.1—we extend the analysis from aggregate

daily generation to hourly operational flexibility. The spot market increases the value of flexible operation, enabling thermal units to shift from rigid baseload behavior toward load-following roles.

We evaluate these effects along both the intensive margin (hourly ramping) and the extensive margin (daily cycling frequency). Table 5 presents the estimates. Columns (1) through (3) use PPML to estimate absolute ramping, ramp-ups, and ramp-downs. The coefficients are positive and statistically significant at the 1% level. For example, the PPML estimate for absolute hourly ramping is 1.382 (Column (1)), implying a $[\exp(1.382) - 1] \approx 298\%$ increase in average absolute hourly ramping. This large semi-elasticity primarily reflects the shift from near-constant baseload output to active intraday load-following: when the pre-reform ramp rate is close to zero for most hours, a discrete transition to regular cycling generates a large proportional change even if absolute ramp magnitudes remain moderate.

Columns (4) through (6) use linear fixed-effects models to study the extensive margin through daily cycling frequency. The reform increases total daily cycling events by 0.073, with corresponding increases in daily startups (0.030) and shutdowns (0.044). These estimates suggest that thermal units shut down more often during low-price periods and restart when residual demand strengthens.

Table 5: Thermal Power Operational Flexibility (90-Day Window)

Dependent Variable:	(1) Absolute Ramping	(2) Ramp Up	(3) Ramp Down	(4) Total Cycling	(5) Startups	(6) Shutdowns
RD-DD Policy Effect	1.382*** (0.030)	1.363*** (0.036)	1.338*** (0.030)	0.073*** (0.015)	0.029*** (0.005)	0.044*** (0.012)
Observations	1,100,520	1,097,654	1,100,916	46,614	46,614	46,614
R^2				0.179	0.130	0.131
Pseudo R^2	0.445	0.269	0.290			
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Columns 1–3 estimate the intensive margin (hourly ramping) using the PPML estimator. Columns 4–6 estimate the extensive margin (daily cycling counts) using a linear fixed-effects model. The models use the RD-DD framework with 364-day alignment. Regressions include quadratic time trend interactions. Standard errors, clustered at the daily level, are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

7.5.3 Intraday Heterogeneity (The “Duck Curve”)

The “duck curve” describes the characteristic intraday shape of net load (total electricity demand minus variable renewable generation) in solar-penetrated power systems: net load is elevated in the morning, drops sharply through midday as solar generation rises, and climbs steeply again in the early evening as solar output falls

while overall demand remains elevated. This shape imposes two sequential operational challenges on the thermal fleet—a rapid midday ramp-down followed by a steep evening ramp-up—both of which require flexibility that coal units designed for baseload operation are poorly equipped to deliver. The hourly analysis tests whether the spot market reform delivers this flexibility: if high-frequency price signals are working as intended, the thermal fleet should be suppressed during solar-peak hours and recover as solar declines, while renewable generation should expand during each technology’s natural resource window.

Figure 4 confirms these predictions across all three panels. The top panel shows that the estimated reduction in thermal generation is most pronounced during midday hours (approximately hours 10 to 16), when solar irradiance is at its daily peak. This midday thermal trough is the operational signature of the duck curve: spot prices fall as solar supply rises, and thermal generators reduce output in response. Through the morning hours, before solar has achieved scale, the thermal reduction is more modest; through the evening hours, as solar output fades and demand remains elevated, the thermal coefficients moderate and move back toward zero, consistent with thermal units ramping up to close the net-load gap. The intraday profile of the thermal response therefore traces the duck curve in reverse—the periods of deepest thermal suppression align precisely with the periods of greatest solar abundance.

The middle panel reveals a complementary temporal profile for wind power. Wind generation gains are distributed across a wider range of hours, with notable increases concentrated during periods of stronger wind resource availability. In the inland plateau geography of this northern Chinese province, wind resources tend to be more abundant during overnight and early-morning hours, and the estimated gains in wind output reflect this diurnal pattern. The positive coefficients during these low-solar hours suggest that the spot market creates operating room for wind absorption precisely when solar is absent: market-based pricing during night and early morning depresses the value of inflexible thermal commitments and allows wind to displace thermal output without competing with solar for the same dispatchable margin.

The bottom panel shows that solar generation gains are concentrated in the daylight window, with the largest coefficients aligning closely with peak irradiance hours. The strong correspondence between the hours of solar gain and the hours of thermal suppression in the top panel provides direct hourly confirmation of the crowding-out mechanism documented at the daily level in Section 5.5.1: the spot market’s downward pressure on thermal dispatch is most acute when solar supply is most abundant, and the two sources are substitutes over the intraday cycle.

Taken together, the three panels trace a coherent picture of intraday net-load reshaping under spot-market pricing. The reform does not simply reduce average thermal output across the clock—it reshapes *when* thermal generation occurs, aligning the oper-

ating profile of the thermal fleet with the intraday pattern of renewable supply. Thermal units provide baseload coverage in the early morning, reduce output during the solar peak, and increase dispatch in the evening as solar fades and wind resources evolve. This intraday dimension of the reform effect, which is invisible in daily aggregate regressions, provides the clearest direct evidence that high-frequency price signals are performing the load-following function that a well-designed spot market is intended to deliver.

7.6 Heterogeneity: Capacity, Age, and Margins

We now quantify how spot-market-induced heterogeneity manifests empirically across operational margins and unit characteristics, drawing on the split-sample analyses described in Section 5.5.

7.6.1 Margins of Adjustment

We first decompose the total reduction in thermal generation into the *extensive margin* (the probability of being dispatched) and the *intensive margin* (the generation level, conditional on operating).

Table 6 reports this decomposition. Column (1) reiterates the total unconditional policy effect (-0.127). Column (2) shows a large and statistically significant reduction in the extensive margin (-0.430 ; $\exp(-0.430) - 1 \approx -35\%$ change in operating probability). This suggests that the market reform reduced the probability that thermal units were dispatched, with larger declines during periods of low spot prices, typically when renewable output was high.

Interestingly, Column (3) shows that the intensive margin is significantly positive (0.299). This indicates that while thermal units are dispatched less frequently, they operate at higher output levels when they are dispatched. Economically, this pattern is consistent with thermal units shifting from continuous baseload operation toward more targeted peaking roles, generating at higher output levels when residual demand is high and spot prices are stronger. The aggregate crowding-out effect is therefore driven by the extensive margin.

7.6.2 Heterogeneity by Unit Characteristics

To examine whether the policy effect differs by unit size and age, we conduct split-sample analyses based on installed capacity and commissioning age.

Installed Capacity (Economies of Scale vs. Flexibility). Table 7 stratifies the sample into Small (< 300 MW), Medium ($300 - 600$ MW), and Large (≥ 600 MW) units.

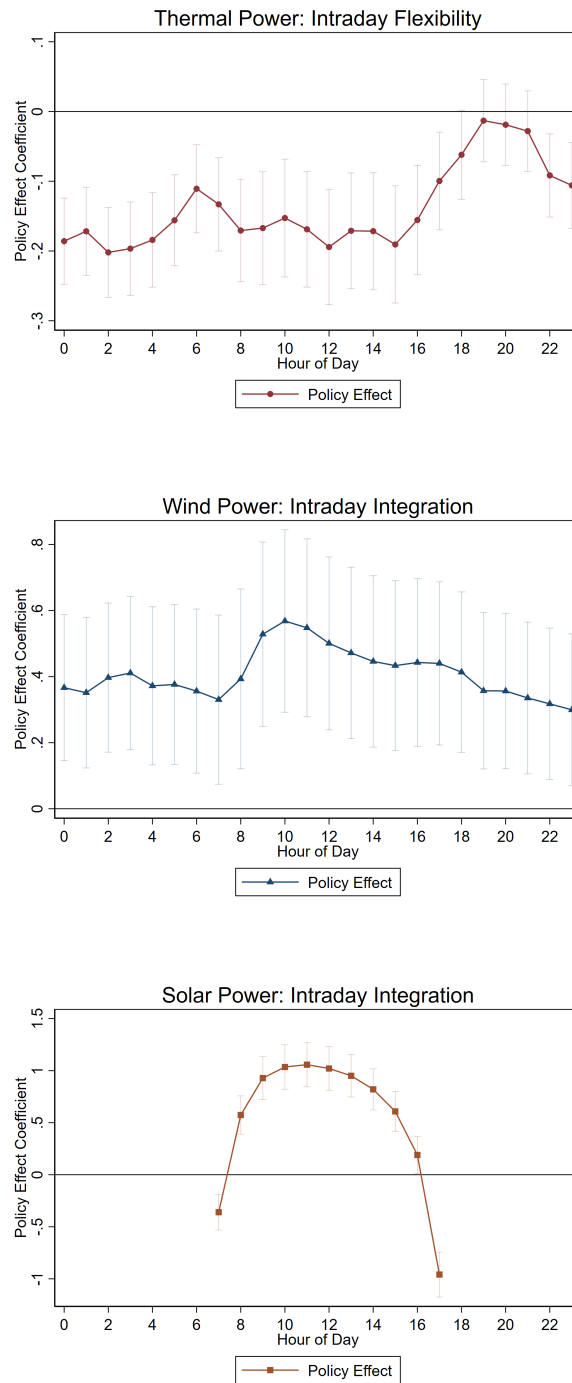


Figure 4: Intraday Integration and Flexibility (Hourly Treatment Effects)

Notes: This figure plots the estimated policy coefficients and 95% confidence intervals from 24 separate hourly regressions for thermal (top), wind (middle), and solar (bottom) power. The models are estimated using the PPML framework, tracking the dynamic “intraday net-load reshaping” response across the 24-hour operational cycle.

Table 6: RD-DD Marginal Effects Decomposition (90-Day Window)

Dependent Variable:	(1)	(2)	(3)
	Total Effect	Extensive Margin	Intensive Margin
RD-DD Policy Effect	-0.127*** (0.027)	-0.430*** (0.016)	0.299*** (0.019)
Units	131	131	131
Observations	45,652	45,652	41,547
Pseudo R^2	0.728	0.028	0.918
Controls	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes

Notes: The table decomposes the overall policy effect using the PPML estimator. The extensive margin utilizes a binary operational indicator, while the intensive margin restricts the sample to observations with positive generation. Standard errors, clustered at the daily level, are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Large units (Panel C) exhibit the largest overall generation reduction (-0.152) and the largest decline in the probability of remaining online (Extensive Margin = -0.674 ; $\exp(-0.674) - 1 \approx -49\%$). By contrast, Small units (Panel A) show a net *increase* in total generation (0.100), reflecting a positive intensive-margin response (0.397).

This pattern is consistent with the merit-order prediction: large coal units typically face high startup costs and rigid minimum-load constraints relative to their operating margin. When renewable output depresses intraday spot prices, these units are displaced at the extensive margin rather than cycling rapidly. Smaller units, with lower startup costs and greater ramping flexibility, are better positioned to start during high-price hours, which accounts for their net increase in conditional output.

Unit Age (Commissioning Age). Table 8 divides the thermal fleet based on the median commissioning age. Older units (Panel B) show a slightly larger overall reduction in generation (-0.134) than younger units (-0.127). The extensive-margin decline is also larger for older units (-0.464 vs. -0.392). This pattern partly reflects the cost-structure heterogeneity documented in the structural analysis: older units tend to have higher marginal costs, placing them higher in the merit order and making them more susceptible to displacement under market-based dispatch. However, as the structural estimates show, startup costs do not follow a simple age gradient—some younger ultra-supercritical units face higher cycling penalties than smaller, older units.

Table 7: Heterogeneity by Unit Capacity (90-Day Window)

Dependent Variable:	(1) Total Effect	(2) Extensive Margin	(3) Intensive Margin
Panel A: Small Units (< 300 MW)			
RD-DD Policy Effect	0.100*** (0.037)	-0.419*** (0.025)	0.397*** (0.017)
Observations	10,798	10,798	9,560
Pseudo R^2	0.683	0.044	0.922
Panel B: Medium Units (300 – 600 MW)			
RD-DD Policy Effect	-0.115*** (0.028)	-0.328*** (0.017)	0.204*** (0.019)
Observations	24,414	24,414	22,570
Pseudo R^2	0.463	0.020	0.749
Panel C: Large Units ($\geq$ 600 MW)			
RD-DD Policy Effect	-0.152*** (0.029)	-0.674*** (0.017)	0.423*** (0.025)
Observations	10,440	10,440	9,417
Pseudo R^2	0.625	0.032	0.840
Controls	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes

Notes: Standard errors, clustered at the daily level, are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Heterogeneity by Unit Age (90-Day Window)

Dependent Variable:	(1) Total Effect	(2) Extensive Margin	(3) Intensive Margin
Panel A: Low Age Units (Young)			
RD-DD Policy Effect	-0.127*** (0.023)	-0.392*** (0.013)	0.301*** (0.018)
Observations	21,896	21,896	20,184
Pseudo R^2	0.627	0.022	0.873
Panel B: High Age Units (Old)			
RD-DD Policy Effect	-0.134*** (0.035)	-0.464*** (0.022)	0.295*** (0.022)
Observations	23,756	23,756	21,363
Pseudo R^2	0.776	0.034	0.934
Controls	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes

Notes: The sample is split based on the median age of the thermal units. Standard errors, clustered at the daily level, are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

8 Structural Estimation Results

8.1 Step 1: Residual Demand Elasticity Estimation

The first step of the structural pipeline estimates residual demand elasticities at the unit level using a Nadaraya–Watson kernel regression. For each unit i and each 15-minute observation (q_{it}, p_{it}) , we compute the local slope of the quantity-price relationship via a Nadaraya–Watson kernel estimator weighted by distance to each evaluation point. We adopt a global bandwidth of $h = 80$ RMB/MWh, selected by a rule-of-thumb cross-validation procedure (see Appendix Table 19), and pool all three sample years to maximize estimation precision.

The estimation sample comprises 9,615,360 unit-period observations. Table 9 summarizes the estimated elasticity distribution across the three estimation approaches: (Method A–Unit) unit-level kernel regression, (Method A–Firm) firm-level kernel aggregation, and (Method B) a Panel IV benchmark. For the main kernel estimate (Method A–Unit), the median residual demand elasticity is -36.35 MWh per RMB/MWh, with a bootstrap 95% confidence interval of $[-37.27, -35.24]$ across 50 resampling iterations.⁶ The mean is substantially more negative (-328.84) due to heavy left-skewness driven by a small number of units with near-flat residual demand at their operating points; the median is therefore the preferred summary statistic. The firm-level kernel (Method A–Firm) yields a median of -14.54 , reflecting aggregation across units within the same parent firm. The Panel IV benchmark (Method B) yields near-zero medians with an extremely volatile mean (34,185), indicating that linear IV at the unit-period level is poorly identified and confirms the value of the nonparametric approach.

Table 9: Residual Demand Elasticity: Cross-Method Comparison

Method	Bandwidth	N	Median	Mean
Method A – Unit Kernel	80	9,615,360	-36.35	-328.84
Method A – Firm Kernel	80	9,615,360	-14.54	-109.96
Method B – Panel IV	—	9,615,360	0.38	34,185.5

Notes: Method A uses Nadaraya–Watson kernel regression at global bandwidth $h = 80$. Method B uses a linear panel IV specification with temperature and wind speed as instruments; the near-zero median and extreme mean indicate poor identification at the unit-period level. Bootstrap 95% CI for the Method A–Unit median across 50 replications: $[-37.27, -35.24]$. Bandwidth robustness results are reported in Appendix Table 19.

These estimated elasticities serve as the key input to the marginal cost recovery step: under the first-order condition for profit maximization, each unit’s MC is identified from its observed price, quantity, and the local slope of its residual demand curve.

⁶The bootstrap resamples units with replacement and re-estimates the full kernel slope for each replication, ensuring that the confidence interval reflects across-unit sampling variation rather than within-unit noise.

8.2 Step 2: Marginal Cost Recovery and Market Conduct

Using the nonparametrically estimated residual demand elasticities from Step 1, we recover the unobserved physical marginal costs (MC) and test the prevailing market conduct among generators. By applying the generalized pricing rule derived from the profit-maximization first-order condition (FOC), we recover the MC curves under three competing behavioral assumptions: Perfect Competition (PC), Unit Cournot-Nash, and Firm Collusive Cournot.

To avoid spurious curvature for units with limited output variation, we employ the sequential “testing down” procedure. The estimation results indicate substantial technological heterogeneity across the thermal fleet. Among the estimated units, the selected functional forms of the marginal cost curves vary with unit operational characteristics. Specifically, 44.9% of the units are best characterized by a constant marginal cost model, 25.4% exhibit a linear cost structure, and 29.7% show significant quadratic costs. This heterogeneity is consistent with differing marginal penalties when units operate near maximum capacity or cycle more frequently.

Under the baseline Perfect Competition assumption, the recovered average marginal costs range predominantly between 332 and 528 RMB/MWh, with a sample mean of approximately 446 RMB/MWh. These estimates are consistent with prevailing coal prices and operating limits in the provincial market during the sample period.

A note on instrument relevance is warranted. The MC recovery equation uses temperature and wind speed as instruments for the residual demand slope. First-stage F-statistics vary across units, with some units showing weaker instrument relevance in the full-year sample. The key specification check is whether the chosen instruments produce economically plausible MC estimates: the recovered distribution (range 332–528 RMB/MWh, mean 446 RMB/MWh) is consistent with prevailing coal prices and operating conditions in the provincial market, and closely matches engineering-based benchmarks for the thermal fleet. Alternative specifications using only the heating-season-excluded subsample produced extreme outliers (unit-level MC up to 1,163 RMB/MWh), confirming that the full-year sample with both instruments is the preferred and more stable choice. We therefore proceed with the full-year estimates; the complete unit-level MC catalog is available upon request.

To systematically identify the competition mechanism, we conduct pairwise model comparisons using the Vuong (1989) test based on the Kullback–Leibler Information Criterion (KLIC). The test results evaluate whether generators exercise unilateral or firm-level coordinated market power. Table 10 summarizes the distribution of the estimated marginal costs and the Vuong test outcomes.

To examine the market structure relevant for these tests, we compare medium-to-long-term (MLT) forward contracts with day-ahead spot transactions. Descriptive

statistics of the operational data show that thermal units are heavily contracted in the forward market. Provincial MLT traded volume is positive and sizable, with a mean of 31.82 MWh per 15-minute period (median: 28.51 MWh, computed from the structural estimation sample). By contrast, day-ahead spot transactions appear to serve mainly as a residual balancing mechanism. Day-ahead traded volumes have a median of 0 and a slightly negative mean of -0.98 MWh, indicating that generators use the spot market primarily for marginal adjustments rather than bulk energy sales. This dominance of MLT contracts over spot trading reshapes generators' spot-market

Table 10: Summary of Marginal Cost Recovery and Vuong Tests

Panel A: MC Functional Form Distribution				
Model Specification	Constant		Linear	Quadratic
Share of Units (%)	44.9%		25.4%	29.7%
Panel B: Average Marginal Cost (RMB/MWh)				
Behavioral Assumption	Mean	Std. Dev.	Min	Max
Perfect Competition (PC)	445.7	36.9	332.3	527.6
Unit Cournot	441.8	38.5	328.9	549.6
Firm Collusive Cournot	442.5	37.6	327.2	536.1
Panel C: Rivers–Vuong Model Selection Test Outcomes				
Test Pair	Model 1 Preferred		Tie	Model 2 Preferred
(1) PC vs. (2) Unit Cournot	87.3%		10.2%	2.5%
(1) PC vs. (2) Firm Cournot	89.8%		9.3%	0.8%

Notes: This table summarizes the recovered marginal cost distributions and market conduct tests based on the structural estimation. Panel A reports the functional form of the MC curves selected by the sequential testing-down procedure. Panel B presents the descriptive statistics of the unit-level average marginal costs under different behavioral assumptions. Panel C reports the share of Vuong test outcomes across units. A model is strictly preferred if the likelihood ratio Z-statistic exceeds the 95% confidence threshold ($|Z| > 1.96$).

incentives—a pattern consistent with the literature on forward contracting and market power in electricity markets. For example, Newbery (1998) shows theoretically that forward contracts reduce generators' incentives to withhold capacity, while Wolak (2007) provides empirical evidence that extensive forward contracting can substantially reduce the incentive to exercise unilateral market power in the spot market, because a large share of output has already been contracted at predetermined prices.

Our Vuong test outcomes are consistent with this interpretation. As shown in Panel C of Table 10, the PC model is significantly preferred to the Unit Cournot model in 87.3% of the tests and is also preferred over the Firm Collusive Cournot model in 89.8% of the tests. This pattern differs from findings for some early deregulated electricity markets. For example, Wolfram (1999), Sweeting (2007), and Puller (2007) document markups or strategic withholding in other settings. One possible interpretation is that

the extensive forward-contract coverage in the province limited the scope for such behavior in the spot market.

Furthermore, this high-contracting market structure helps explain a quantitative regularity in Panel B of Table 10: the average MC recovered under the PC model (445.7 RMB/MWh) is very close to the MCs recovered under the Unit Cournot (441.8 RMB/MWh) and Firm Cournot (442.5 RMB/MWh) models, with differences below 1%. Under the Cournot assumptions, whenever a generator is a net seller in the spot market (i.e., generation exceeds its contract position), the model rationalizes this exposure through a positive markup ($P > MC$). The implied marginal cost is therefore lower because part of the observed price is attributed to market power. Under the PC assumption, observed prices are attributed entirely to production costs ($P = MC$), yielding a mean recovered MC of 445.7 RMB/MWh—less than 1% above the Cournot estimates—indicating that forward-contract coverage effectively eliminates the scope for large Cournot markups in this market.

Taken together, the Vuong tests and the institutional evidence are consistent with a spot market that behaves approximately competitively. We therefore use the Perfect Competition assumption in the subsequent dynamic modeling of startup costs and counterfactual dispatch.

8.3 Heterogeneity in Marginal Costs and Market Merit Order

Given the evidence in favor of the Perfect Competition assumption, we next examine heterogeneity in recovered marginal costs across the thermal fleet. The resulting merit-order curves provide a structural benchmark for interpreting the heterogeneous generation responses in the RD-DD estimates.

8.3.1 Cost Structure and Economies of Scale (Installed Capacity)

We first relate recovered marginal costs to installed capacity. Figure 5 shows the group-level MC curves by nameplate capacity. Small units (< 300 MW) have steeper marginal cost curves and reach high cost levels at relatively low cumulative capacity. Medium (300–600 MW) and large (≥ 600 MW) units instead exhibit flatter cost schedules over a wider output range.

Figure 6 places these units in the aggregate merit order. Small units are concentrated in the high-cost right tail of the supply curve, often above 475 RMB/MWh, whereas medium and large units account for much of the flatter middle and lower portions.

This pattern helps interpret the reduced-form heterogeneity results in Table 7. Because small units are concentrated at the upper end of the merit order, they are less likely to serve as baseload units under competitive dispatch. Instead, they are dispatched mainly in high-price periods. This interpretation is consistent with the positive

intensive-margin response for small units in the RD-DD estimates.

By contrast, larger baseload units occupy lower positions in the merit order but face greater operational rigidities. As the next section shows, their adjustment depends not only on marginal fuel cost but also on startup costs and operational flexibility when within-day prices fall.

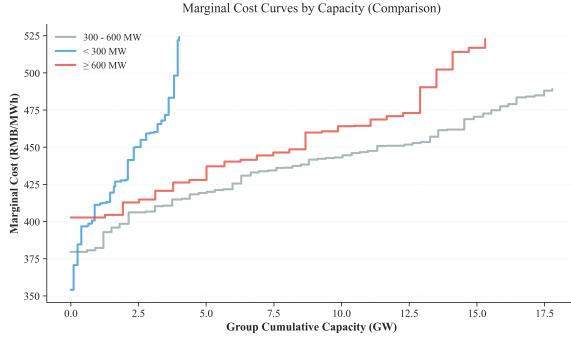


Figure 5: Marginal Cost Curves by Capacity

Notes: Figure 5 plots the recovered MC curves aggregated by capacity tiers. Figure 6 maps the units onto the global merit order curve, highlighting the concentration of small units (< 300 MW) at the high-cost right tail.

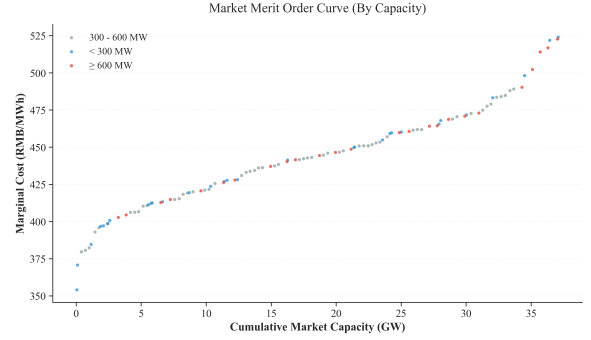


Figure 6: Market Merit Order Curve (By Capacity)

8.3.2 Cost Structure and Unit Age

We next examine how marginal costs vary with unit vintage. Figure 7 presents group-level marginal cost curves for older units (above the median age) and younger units (below the median age). The estimates indicate that older units tend to have higher marginal costs across much of the capacity range, consistent with lower thermal efficiency in older plants.

Figure 8 shows the corresponding aggregate merit order. Older units are more concentrated in the higher-cost right tail of the distribution.

These structural patterns help explain the reduced-form results in Table 8. If dispatch follows the merit order, units in the right tail are more exposed to displacement when residual demand falls, as occurs during periods of strong renewable generation. This interpretation is consistent with the finding that older units experience a larger extensive-margin reduction than younger units.

8.4 Step 3: Dynamic State Transitions and Startup Cost Estimation

Before estimating startup costs with the B-spline Probit and Simulated Minimum Distance (SMD) framework, we specify how generators form expectations about future market conditions. The two continuous state variables in the structural model are aggregate thermal generation (L_t , scaled by 10,000 for numerical stability) and the spot

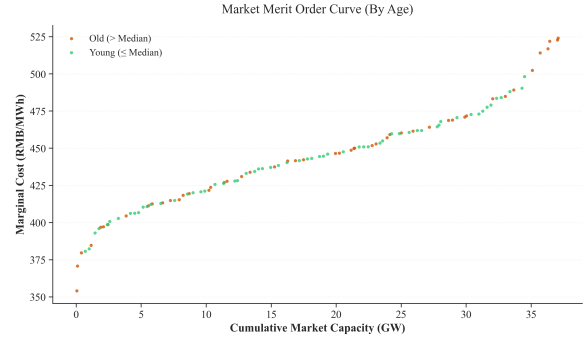
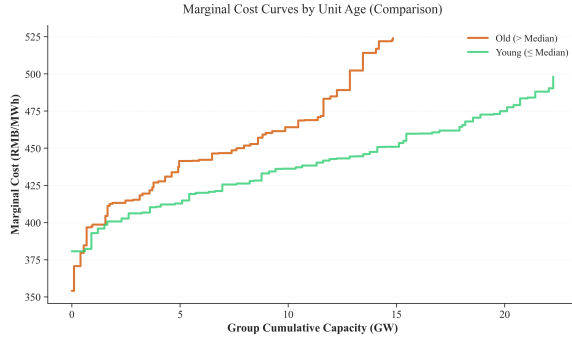


Figure 7: Marginal Cost Curves by Unit Age Figure 8: Market Merit Order Curve (By Age)
Notes: Figure 7 plots the recovered MC curves aggregated by age group. Figure 8 displays the global merit order supply curve, with each dot representing a unit's average marginal cost evaluated at its optimal dispatch capacity, colored by vintage.

price (P_t). Load follows a first-order autoregressive (AR(1)) process estimated by OLS. For the price process, we adopt a two-regime Markov regime-switching specification as the primary model, which better captures the spike dynamics and bimodal distribution of intraday electricity prices; an AR(1) price model is estimated as a robustness check. Both equations include weather controls and hour-of-day and month-of-year fixed effects to capture predictable diurnal and seasonal patterns.

Table 11 reports the OLS estimates of the load AR(1) and the AR(1) price robustness specification. The Markov-switching price parameters (regime means, variances, and transition probabilities) are estimated by maximum likelihood. Key estimates include two regimes with distinct mean prices (a low-price baseload regime and a high-price peak regime) and regime-specific persistence; full parameter estimates are provided in the Online Appendix.

The estimates reveal two key structural features. First, aggregate thermal generation exhibits moderate persistence ($\rho_L = 0.7146$), reflecting the gradual adjustment of the committed thermal fleet within the day. Second, in the AR(1) robustness specification, the spot price displays stronger autoregressive memory ($\rho_P = 0.9815$) and is highly correlated with the contemporaneous thermal generation level ($\hat{\beta}_P = 24.6041$), consistent with the pattern that periods of higher aggregate thermal output tend to coincide with higher spot prices. These parameters anchor the forward-looking valuations regardless of which price process is used in the primary simulation.

8.5 Policy Function and Dynamic Value Approximation

Resolving the dynamic optimal stopping problem over a high-frequency, 48-hour forward-looking horizon ($H = 192$ periods) poses the curse of dimensionality. To overcome this, we approximate the forward-looking value functions using a B-spline Probit framework. We deploy an adaptive dimensionality reduction strategy to guarantee

Table 11: Estimation Results of State Transition Models: Load AR(1) (Primary) and Price AR(1) (Robustness)

Dependent Variable: Model:	Thermal Generation (Scaled) Load AR(1)	Spot Price (RMB/MWh) Price AR(1) (Robustness)
Lagged Load (L_{t-1})	0.7146	–
Lagged Price (P_{t-1})	–	0.9815
Contemp. Thermal Gen. (L_t)	–	24.6041
Weather Controls	Yes	Yes
Intercept	0.1734	-8.2158
Month Fixed Effects	Yes	Yes
Hour Fixed Effects	Yes	Yes
Residual Std. Dev. (σ)	0.0673	48.1805

Notes: This table reports OLS estimates of the AR(1) state transition models. Thermal generation is the sum of observed thermal dispatch, scaled by 10,000 for numerical stability. Weather controls (local temperature and wind speed) are included in both equations but are small in magnitude relative to the persistence terms. The residual standard deviations (σ) are used to simulate forward-looking stochastic shocks in the AR(1) robustness specification of the dynamic programming solver; the primary specification uses Markov regime-switching price parameters. The coefficient on contemporaneous thermal generation ($\hat{\beta}_p = 24.6041$) in the price equation is estimated by OLS and should be interpreted as a reduced-form correlation rather than a structural causal effect, since aggregate thermal generation and the spot price are simultaneously determined within each dispatch period.

global convergence, particularly for peaker units with sparse operating histories. The algorithm initializes with a flexible third-degree polynomial using 5 quantile-based knots, adaptively falling back to lower-degree specifications if sample sparsity induces matrix singularity.

Table 12 summarizes the average estimated coefficients of the policy functions for the 114 thermal units included in the estimation sample. The key regressors are the lagged operating-status indicator (d_{it-1}), the B-spline basis expansion of the unit’s potential operating profit (π_{it} , evaluated at the current state), and the basis expansion of scaled aggregate thermal generation (L_t).

The policy function estimates highlight the main drivers of generator commitment decisions. In particular, the lagged operating-status indicator (d_{it-1}) has a large positive mean coefficient of 4.303, consistent with the role of startup costs in creating persistence in operating decisions: units already online are substantially less likely to shut down within the period. Aggregate thermal generation enters the policy function non-linearly via B-spline basis functions ($\psi_k(L_t)$), with the first two basis coefficients of opposite sign reflecting the non-monotone relationship between aggregate load and individual shutdown probabilities during periods of depressed residual demand, such as midday intervals with high solar generation.

Table 12: Average Estimated Coefficients of the Policy Function (B-spline Probit)

Variables	Mean Coefficient
State Dependence	
Lagged Operating Status (d_{it-1})	4.3033
Intercept	7.0673
Potential Profit Splines ($\phi_k(\pi_{it})$)	
Basis function ϕ_0	-2.1774
Basis function ϕ_1	-2.4606
Basis function ϕ_2	-2.3621
Basis function ϕ_3	-2.5082
Basis function ϕ_4	-1.5185
Basis function ϕ_5	-1.8728
Thermal Generation Splines ($\psi_k(L_t)$)	
Basis function ψ_0	-742.0826
Basis function ψ_1	604.1973
Basis function ψ_2	-20.1395
Basis function ψ_3	-7.4337
Basis function ψ_4	-10.3059
Basis function ψ_5	-9.1316
Month Fixed Effects	Yes
Hour Fixed Effects	Yes

Notes: This table reports the average coefficients of the B-spline Probit models across all 114 units. The dependent variable is the binary operating indicator ($d_{it} \in \{0, 1\}$). Basis functions ϕ_k and ψ_k are cubic B-spline bases evaluated at the potential profit π_{it} and scaled aggregate thermal generation L_t , respectively. Because the knot locations and basis support are calibrated separately for each unit, the reported averages summarize the direction and approximate magnitude of each component but are not directly comparable across individual units.

8.5.1 Structural Recovery of Unit-Level Startup Costs ($H = 192$ Horizon)

Using the estimated state-transition parameters and the non-linear policy functions, we simulate $N = 500$ forward paths over the $H = 192$ quarter-hour periods (a 48-hour forward horizon, our primary specification; see Appendix Table 20 for robustness across $H \in \{96, 192, 384\}$) to evaluate the expected discounted value gain from starting relative to waiting. We then recover the latent startup costs (SC_i) via a Simulated Minimum Distance (SMD) estimator implemented through Weighted Non-Linear Least Squares (WNLLS).

To balance idiosyncratic operational heterogeneity with numerical robustness, we use a two-branch estimation strategy. Among the 114 thermal units, 85 units have sufficient high-frequency cycling behavior ($N_{trans} \geq 30$)⁷ to be identified via Individual Estimation (Branch A). The remaining 29 rigid baseload units are estimated using Pooled Estimation (Branch B), where moments are pooled within capacity tiers. Appendix Table 21 presents the structural estimation results for the pooled parameters.

The pooled parameters summarize group-level cost structure, but the pooled estimates alone would miss the substantial within-group heterogeneity in startup costs. To characterize the fleet’s operational frictions without presenting unit-by-unit outputs in the main text (the full unit-level catalog of startup costs is available upon request), we construct the unified distribution of final recovered unit-specific startup costs (SC_i) by combining both estimation branches. As noted in Section 6 (footnote therein), 114 of the 162 structural-estimation units converge to stable cost estimates and are used in the counterfactual simulations. Table 13 presents descriptive statistics for the recovered startup costs across the three capacity subgroups.

Table 13: Descriptive Statistics of Structurally Recovered Unit-Level Startup Costs ($H = 192$ Horizon)

Capacity Group	N	Mean (RMB)	Median (RMB)	Std. Dev.	Minimum	Maximum
Small Units (< 300 MW)	31	93,221	43,877	104,312	9,747	401,546
Medium Units (300–600 MW)	58	139,958	62,678	159,015	17,741	657,530
Large Units (≥ 600 MW)	25	316,163	170,315	323,856	16,986	1,117,608
Total Fleet	114	165,890	65,759	211,446	9,747	1,117,608

Notes: This table reports the descriptive statistics of the final recovered latent startup costs (SC_i) for all 114 thermal units under the primary $H = 192$ quarter-hour simulation horizon (48-hour forward window). Units are classified into capacity tiers consistent with the pooled estimation groups. All monetary values are expressed in RMB. Sensitivity estimates for $H = 96$ and $H = 384$ horizons are presented in Appendix Table 20; the $H = 192$ and $H = 384$ distributions are nearly identical, confirming robustness to the horizon choice.

The distribution of recovered startup costs shows substantial economies of scale,

⁷The threshold of $N_{trans} \geq 30$ is chosen to ensure that the B-spline Probit likelihood is well-identified at the unit level: with fewer than 30 observed on/off transitions, the spline basis tends to be poorly supported and the Hessian becomes near-singular. Units below this threshold are assigned to Branch B, where moments are pooled within capacity tiers to stabilize estimation.

together with considerable within-group dispersion. Mean startup cost rises from 93,221 RMB for Small units to 139,958 RMB for Medium units and 316,163 RMB for Large units. The median follows a similar gradient: 43,877 RMB (Small), 62,678 RMB (Medium), and 170,315 RMB (Large).

The distributions within each group are also right-skewed, as the means substantially exceed the corresponding medians and the standard deviations are large. For example, the median Large unit has a startup cost of 170,315 RMB, while the maximum in this group reaches 1,117,608 RMB (≈ 1.12 million RMB). This dispersion suggests that some older baseload units face particularly high mechanical and thermal costs when cycling. The total-fleet median (65,759 RMB) is the primary summary statistic used in the counterfactual simulations.

These structurally recovered parameters help interpret the reduced-form findings by unit capacity. The relatively high startup costs of Large units (≥ 600 MW) are consistent with the substantial extensive-margin decline in the RD-DD estimates. When renewable output depresses intraday prices below their marginal costs, these units are less likely to cycle and re-enter quickly. By contrast, Small units (< 300 MW), with much lower average startup costs, are more able to start during high-price hours, which is consistent with their stronger intensive-margin response.

These recovered startup-cost estimates provide the structural basis for the counterfactual economic dispatch and unit-commitment simulations used in the final decomposition of renewable integration, engineering cost, and market-accounting incidence.

8.6 Joint Distribution of Operational Costs and Reduced-Form Synthesis

To synthesize the structural estimates and summarize the operational constraints facing the thermal fleet, we analyze the joint distribution of recovered average marginal costs and startup costs. We link these structural parameters to unit characteristics to explain the heterogeneous behavioral responses identified in the reduced-form RD-DD analysis.

Figure 9 illustrates the overall kernel density estimates for both cost components across the evaluated fleet. The overall average marginal cost (Panel a) exhibits a relatively concentrated distribution with a mean of 446 RMB/MWh (range: 332–528 RMB/MWh), reflecting the relatively homogeneous coal fuel base across the fleet, with modest right skew for less efficient peaking units. In contrast, the overall startup cost distribution (Panel b) is strongly right-skewed. While the mean startup cost is 165,890 RMB per start, much of the distribution is concentrated at the lower end (median 65,759 RMB), with a long right tail reaching 1,117,608 RMB for the most rigid units. This right tail is consistent with large discrete operating frictions for some rigid baseload

units, for which cycling entails substantial mechanical wear and thermal degradation.

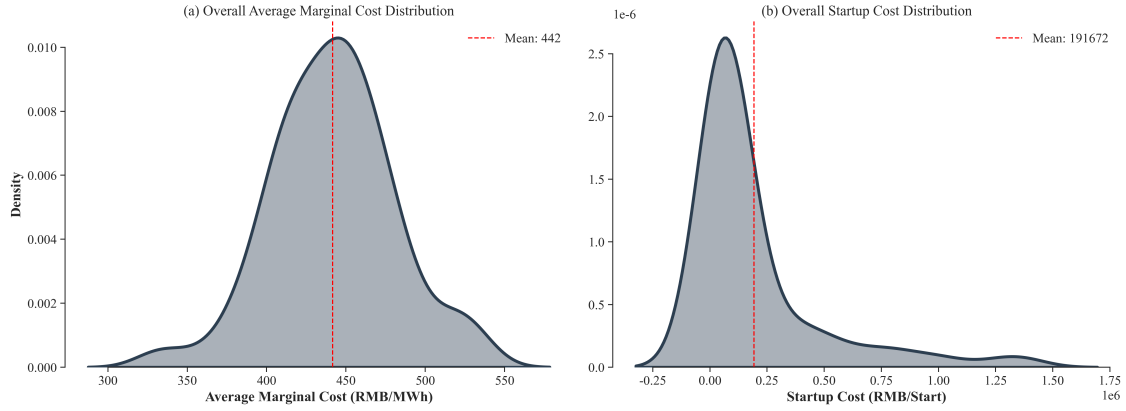


Figure 9: Overall Distribution of Average Marginal Costs and Startup Costs

Notes: This figure plots the kernel density estimates for the structurally recovered average marginal costs (left) and startup costs (right) across the entire thermal fleet. The dashed red lines indicate the sample means.

To unpack this variation, we first decompose these distributions by installed capacity. Figure 10 stratifies the cost distributions by unit size, which relates directly to the economies-of-scale pattern in Table 13. The startup cost distribution (Panel b) shows that Small units (< 300 MW, in blue) are concentrated near the lower end of startup costs. As capacity increases, the distribution shifts to the right and becomes more dispersed. Large units (≥ 600 MW, in red) therefore tend to face higher startup constraints.

This structural mapping helps interpret the reduced-form capacity heterogeneity. In the RD-DD estimates, large units show the largest extensive-margin decline (-0.674), while small units exhibit a stronger intensive-margin response ($+0.397$). The structural distributions are consistent with this pattern: large baseload units face higher startup costs, which makes cycling less attractive when renewable generation depresses intra-day spot prices. Small units, by contrast, have lower startup costs and are therefore more able to operate during high-price hours.

Furthermore, Figure 11 explores heterogeneity by commissioning age and relates it to the unit-age results in Table 8. When the fleet is stratified by the sample median commissioning age (15 years), the average marginal cost distributions (Panel A) indicate that older units (> 15 years, in blue) tend to occupy a somewhat higher cost range, consistent with physical degradation. Panel B shows that younger units (≤ 15 years, in orange) exhibit a heavier right tail in the startup cost distribution.

These complementary patterns help account for the empirical findings. The RD-DD estimates show that older units experience a larger extensive-margin reduction (-0.464 vs. -0.392). The structural estimates suggest that this difference is mainly related to their higher marginal costs, which place them higher in the merit order and make them more likely to be displaced by renewables. By contrast, younger units are often large,

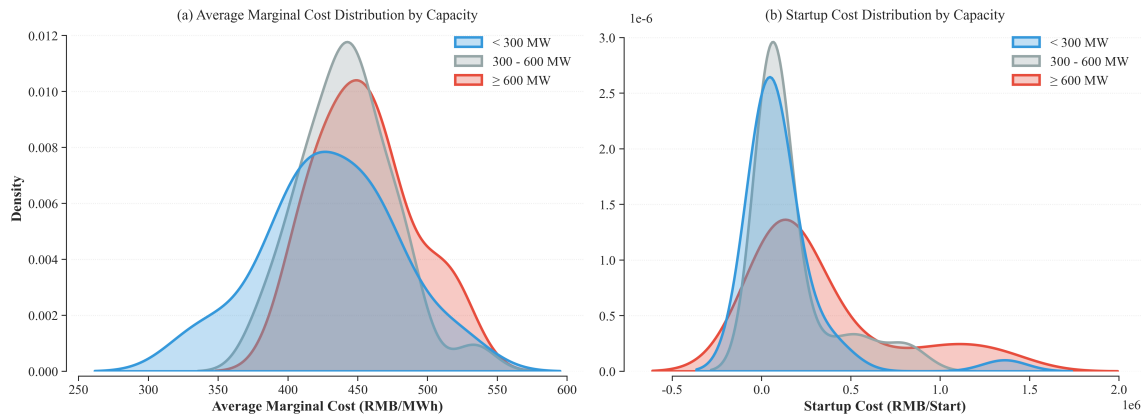


Figure 10: Cost Distributions Stratified by Installed Capacity

Notes: This figure maps the kernel density distributions of average marginal costs and startup costs categorized by nameplate capacity tiers, highlighting the structural inflexibility of larger units.

ultra-supercritical baseload units. Although these units tend to have lower marginal costs, their more complex physical configuration can imply higher startup costs. This combination is consistent with greater reluctance to shut down during temporary low-price periods. This counterintuitive age-startup cost relationship—where newer units face higher cycling penalties—reflects the technological composition of China’s thermal fleet and should not be confused with a general efficiency advantage for older units; the dominant displacement mechanism for older units operates through higher marginal costs, not lower startup costs.

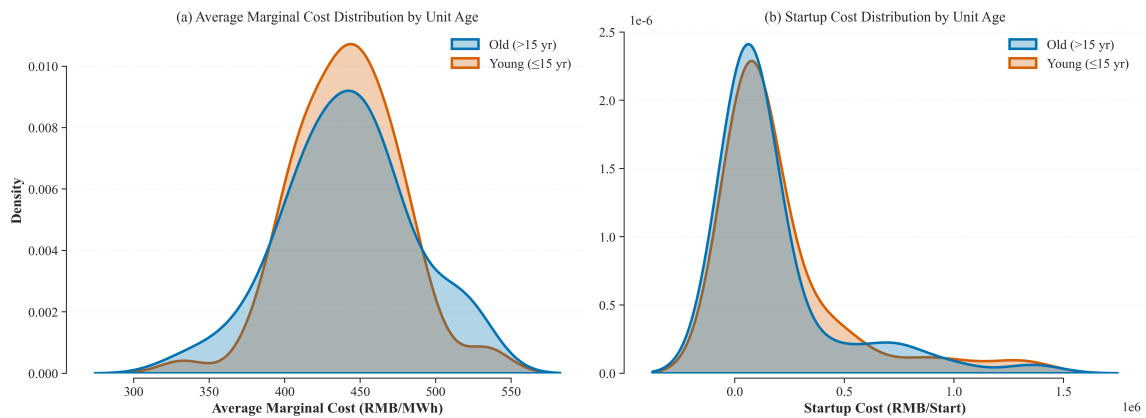


Figure 11: Cost Distributions Stratified by Unit Age

Notes: This figure contrasts the cost distributions between units commissioned more than 15 years ago and younger units (≤ 15 years). It shows that older units tend to have higher marginal costs, while newer units tend to have higher startup frictions.

Together, the distribution of these structural parameters suggests that the spot market reform reallocates generation along multiple cost dimensions. Units with sufficiently low marginal costs are better positioned to remain in baseload operation, while units with sufficiently low startup costs are better positioned to respond to short-run price variation. Units with neither advantage face the greatest exposure to output

displacement under merit-order dispatch.

8.7 Micro-Behavioral Heterogeneity: The Cost Quadrant Framework

To systematically bridge our structural estimates with the reduced-form RD-DD findings, we construct a two-dimensional cost quadrant framework. We classify the thermal fleet into four distinct technological archetypes based on whether their structurally recovered Average Marginal Cost (MC) and Startup Cost (SC) fall above or below their respective sample medians.

8.7.1 Technological Mapping of the Fleet

Figure 12 and Figure 13 map the thermal units onto this 2×2 operational space, categorized by commissioning age and nameplate capacity, respectively. The four resulting quadrants represent fundamentally different physical and economic capabilities:

- **Panel A: Flexible Baseload (Low MC, Low SC).** Units in this group combine relatively low marginal costs with relatively low startup costs, allowing them to remain online while also adjusting more easily to intraday price movements.
- **Panel B: Rigid Baseload (Low MC, High SC).** These units have relatively low fuel costs but face larger thermal stress penalties when cycling.
- **Panel C: Flexible Peaking (High MC, Low SC).** This group is concentrated among older, small-to-medium units. Although they are less competitive in fuel cost, their relatively low startup frictions make them more adaptable.
- **Panel D: Inflexible High-Cost (High MC, High SC).** This group is concentrated among older, medium-to-large units that face both relatively high marginal costs and relatively high cycling frictions.

These distributions are consistent with the underlying drivers of technological heterogeneity: a unit's age primarily determines its horizontal placement (marginal cost efficiency), while its capacity determines its vertical placement (startup cost inflexibility).⁸

⁸This pattern follows from two mechanisms identified in the structural estimates. First, older units exhibit higher marginal costs (Figure 7), consistent with physical degradation of thermal efficiency over time. Second, larger units tend to have higher startup costs (Table 13), reflecting the greater mechanical and thermal stress associated with cycling high-mass boiler and turbine systems.

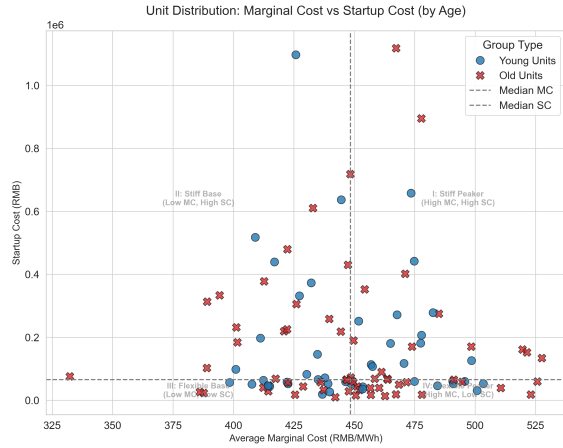


Figure 12: Cost Quadrants by Unit Age
Notes: These scatter plots map the thermal fleet across structurally recovered Marginal Costs (x-axis) and Startup Costs (y-axis). The dashed lines denote the respective sample medians, delineating the four technological archetypes.

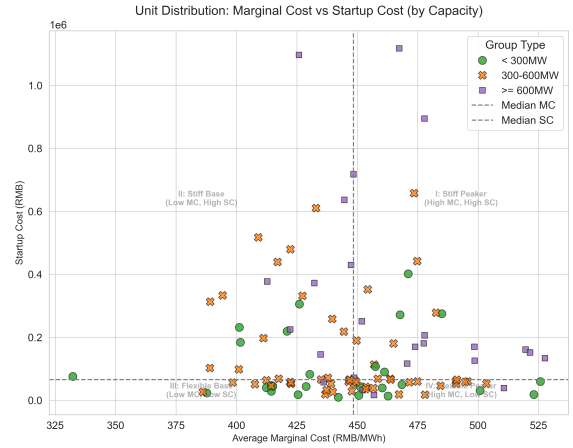


Figure 13: Cost Quadrants by Unit Capacity

8.7.2 Generation Adjustments: The Dual Margins of Cost

To formalize how these structural constraints dictate market behavior, we re-estimate the baseline RD-DD models within each of the four cost quadrants. Table 14 presents the decomposition of the policy effect into the extensive and intensive margins. By systematically cross-comparing these quadrants, we can isolate how generators trade off fuel efficiency against cycling flexibility.

The Impact of Marginal Cost (A vs. C, B vs. D): Holding operational flexibility constant, marginal cost is strongly associated with the magnitude of the generation response. Among units with low startup costs (Panels A and C), the high-MC group shows a larger decline in total generation (-0.327 vs. -0.141) and a larger reduction in the extensive margin (-0.454 vs. -0.338). This pattern is consistent with less fuel-efficient units being more exposed to displacement after the reform. The same contrast appears among high-startup-cost units (Panels B and D). Rigid Baseload units (Panel B) exhibit a statistically insignificant total-generation effect (-0.053 , $p > 0.1$), while their large extensive-margin reduction (-0.432) is almost fully offset by an equally large intensive-margin gain ($+0.341$), suggesting that these units maintain their overall utilization by concentrating output into higher-value hours. Inflexible High-Cost units (Panel D), by contrast, experience a significant decline in total generation (-0.134), with the extensive margin (-0.647 ; $\exp(-0.647) - 1 \approx -48\%$ reduction in operating probability) only partially offset by the intensive margin ($+0.554$; $\exp(0.554) - 1 \approx +74\%$ conditional generation increase). Taken together, these comparisons suggest that lower marginal cost is an important determinant of dispatch under spot pricing, but startup costs can attenuate the total-generation effect by shifting adjustment toward

Table 14: Heterogeneity by Cost Quadrants (90-Day Window)

Dependent Variable:	(1) Total Effect	(2) Extensive Margin	(3) Intensive Margin
Panel A: Flexible Baseload (Low MC, Low SC)			
RD-DD Policy Effect	-0.141*** (0.034)	-0.338*** (0.024)	0.179*** (0.017)
Panel B: Rigid Baseload (Low MC, High SC)			
RD-DD Policy Effect	-0.053 (0.034)	-0.432*** (0.025)	0.341*** (0.021)
Panel C: Flexible Peaking (High MC, Low SC)			
RD-DD Policy Effect	-0.327*** (0.050)	-0.454*** (0.036)	0.118*** (0.024)
Panel D: Inflexible High-Cost (High MC, High SC)			
RD-DD Policy Effect	-0.134*** (0.044)	-0.647*** (0.032)	0.554*** (0.029)
Controls & Fixed Effects	Yes	Yes	Yes

Notes: This table reports PPML estimates of the RD-DD policy effects decomposed by the four structural cost quadrants. Standard errors, clustered at the daily level, are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

within-day output concentration.

The Impact of Startup Cost (A vs. B, C vs. D): Holding marginal cost constant, startup cost is associated with how units adjust along the intensive and extensive margins. Among low-MC units, the high-startup-cost group (Panel B) records a larger intensive-margin increase than the low-startup-cost group (Panel A, +0.341 vs. +0.179), while the total-generation effect falls to statistical insignificance. This is consistent with Rigid Baseload units absorbing the extensive-margin reduction almost entirely through within-day output concentration, rather than an overall drop in utilization. Among high-MC units, the contrast between Panels C and D points in the same direction: Panel D shows a larger intensive-margin gain (+0.554 vs. +0.118) alongside a smaller total decline (−0.134 vs. −0.327), suggesting that high startup costs again push units toward within-day adjustment. Panel D also combines this with a small and statistically insignificant cycling response (Table 15), consistent with startup frictions constraining extensive-margin participation for units that are already costly to run.

8.7.3 Operational Flexibility: Ramping and Cycling Dynamics

Table 15 further examines how cost structure is associated with two margins of operational flexibility: intensive ramping and extensive daily cycling.

Table 15: Heterogeneity of Operational Flexibility by Cost Structure (90-Day Window)

Cost Archetype	(1) Abs Ramp	(2) Ramp Up	(3) Ramp Down	(4) Total Cycling	(5) Startups	(6) Shutdowns
Panel A: Flexible Baseload (Low MC, Low SC)	1.518*** (0.039)	1.491*** (0.044)	1.488*** (0.038)	0.129*** (0.023)	0.058*** (0.013)	0.071*** (0.013)
Panel B: Rigid Baseload (Low MC, High SC)	1.488*** (0.045)	1.493*** (0.048)	1.437*** (0.047)	0.054*** (0.018)	0.025*** (0.009)	0.029** (0.013)
Panel C: Flexible Peaking (High MC, Low SC)	1.312*** (0.052)	1.303*** (0.054)	1.292*** (0.053)	0.072*** (0.014)	0.025*** (0.006)	0.047*** (0.010)
Panel D: Inflexible High-Cost (High MC, High SC)	1.108*** (0.049)	1.081*** (0.050)	1.051*** (0.051)	0.035 (0.025)	0.009 (0.010)	0.026 (0.017)

Notes: This table assesses the RD-DD policy impact on intensive ramping (Columns 1-3, PPML) and extensive daily cycling (Columns 4-6, Linear FE) across the structural cost quadrants. Standard errors are clustered at the daily level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The Ramping vs. Cycling Trade-off (A vs. B, C vs. D): Startup costs are closely related to the balance between continuous load-following and discrete commitment adjustment. Among low-MC units (Panels A and B), the high-startup-cost group cycles substantially less (0.054 vs. 0.129) while exhibiting similarly large ramping responses, consistent with adjusting output within the day rather than frequently shutting down and restarting. Among high-MC units (Panels C and D), the contrast is also pronounced. Flexible Peaking units (Panel C) show a meaningful increase in daily cycling (0.072), whereas the cycling coefficients for Inflexible High-Cost units (Panel D) are small and statistically insignificant (0.035). This pattern suggests that startup frictions are an important constraint on extensive-margin adjustment for high-cost units.

The Baseline Competitiveness Effect (A vs. C, B vs. D): Marginal cost is also associated with the way flexibility is expressed. Comparing Panels A and C, the high-MC group shows less ramping but more cycling, consistent with being dispatched less continuously and instead adjusting more through on-off decisions during higher-value hours. Comparing Panels B and D, the high-MC group exhibits both weaker ramping and little cycling response. Together, these results suggest that marginal cost shapes a unit's baseline likelihood of dispatch, while startup cost shapes whether adjustment occurs mainly through within-day ramping or through discrete commitment changes.

Together, the structural and reduced-form evidence points to a common pattern: units with higher marginal costs are more exposed to displacement under spot pricing, and units with higher startup costs are less able to participate actively in daily balancing.

8.8 Step 4: Counterfactual Simulation: Engineering-Cost Gains, Renewable Integration, and Market Incidence

The counterfactual exercise evaluates five scenarios (S1–S5) relative to observed baseline operations. Scenario Group I (S1–S2) holds the observed gross-demand path fixed and isolates the engineering-cost savings from merit-order redispatch and commitment reoptimization within the thermal fleet. Scenario Group II (S3–S5) keeps gross system demand at the modeled pre-reform level, replaces observed wind and solar generation with the RD-DD-implied counterfactual renewable path, and then solves for thermal dispatch, renewable utilization, curtailment, shortage, and oversupply subject to unit-level operating constraints. Tables 16–18 below present the full set of results. Table 16 reports profits, consumer payment, effective clearing price, and total social surplus—the headline market-incidence outcomes. Table 17 reports generation mix and output reallocation—the physical basis for the incidence results. Table 18 provides the thermal-side revenue and cost decomposition, and Table 18 reports renewable revenue by technology.

Table 16: Counterfactual Simulation: Profits, Consumer Payment, and Effective Clearing Price

Scenario / Comparison	Thermal Profit (bn RMB)	Renewable Profit (bn RMB)	Consumer Payment (bn RMB)	Effective Clearing Price (RMB/MWh)	Total Social Surplus (Δ vs. Baseline, bn RMB)
Panel A: Scenario levels					
1. Baseline (Observed)	-4.775	1.323	39.096	332.000	0.000
2. S1 (Optimal Dispatch)	9.918	3.649	53.951	459.667	+2.164
3. S2 (Optimal Commitment & Dispatch)	10.356	3.494	53.182	452.509	+3.216
4. S3 (Fixed Commitment, Reference Dispatch)	-3.229	1.837	39.096	332.000	+2.060
5. S4 (Fixed Commitment, Economic Redispatch)	8.433	4.368	51.828	444.869	+3.521
6. S5 (Endogenous Commitment & Redispatch)	9.289	4.331	51.630	442.823	+4.538
Panel B: Selected cross-scenario changes					
Baseline $\rightarrow$ S1	14.693	2.326	14.855	127.667	+2.164
S1 $\rightarrow$ S2	0.438	-0.155	-0.769	-7.158	+1.052
Baseline $\rightarrow$ S3	1.546	0.514	0.000	0.000	+2.060
S1 $\rightarrow$ S4	-1.485	0.719	-2.123	-14.798	+1.357
S3 $\rightarrow$ S4	11.662	2.531	12.732	112.869	+1.461
S4 $\rightarrow$ S5	0.856	-0.037	-0.198	-2.046	+1.017
S2 $\rightarrow$ S5	-1.067	0.837	-1.552	-9.686	+1.322
Baseline $\rightarrow$ S5	14.064	3.008	12.534	110.823	+4.538

Notes: This table reports market-accounting incidence. Thermal and renewable profits are accounting profits in the model, not full welfare measures. Consumer payment is the model-implied total payment by consumers under the scenario-specific effective clearing price and served load. Total Social Surplus is computed as $\Delta SS = \Delta \Pi_{th} + \Delta \Pi_{re} - \Delta \text{Payment}$ relative to the Baseline, the standard welfare measure under perfectly inelastic demand. In Panel A, Baseline is normalized to zero and each row reports the cumulative welfare gain from that scenario relative to Baseline. In Panel B, each entry reports the welfare change from the named cross-scenario comparison. All monetary values are expressed in billions of RMB and rounded to three decimals.

8.8.1 Decomposing Engineering-Cost Savings: Intensive versus Extensive Margins

Table 17 establishes the physical benchmark for the counterfactual exercise. Along the observed-load path, S1 and S2 leave the generation mix unchanged relative to the baseline, so the thermal cost savings in Scenario Group I (Table 18, Panel B, rows 1–2) do not come from substituting across technologies. Instead, they arise from a more

Table 17: Counterfactual Simulation: Generation Mix and Output Reallocation

Scenario / Comparison	Thermal Generation (TWh)	Wind Generation (TWh)	Solar Generation (TWh)	Total Generation (TWh)
Panel A: Scenario levels				
1. Baseline (Observed)	98.143	13.916	5.700	117.759
2. S1 (Optimal Dispatch)	98.143	13.916	5.700	117.759
3. S2 (Optimal Commitment & Dispatch)	98.143	13.916	5.700	117.759
4. S3 (Fixed Commitment, Reference Dispatch)	90.376	19.213	8.170	117.759
5. S4 (Fixed Commitment, Economic Redispatch)	90.376	19.213	8.170	117.759
6. S5 (Endogenous Commitment & Redispatch)	90.452	19.214	8.169	117.835
Panel B: Selected cross-scenario changes				
Baseline → S1	0.000	0.000	0.000	0.000
S1 → S2	0.000	0.000	0.000	0.000
Baseline → S3	-7.767	5.297	2.470	0.000
S1 → S4	-7.767	5.297	2.470	0.000
S3 → S4	0.000	0.000	0.000	0.000
S4 → S5	0.076	0.001	-0.001	0.076
S2 → S5	-7.691	5.298	2.469	0.076
Baseline → S5	-7.691	5.298	2.469	0.076

Notes: This table reports the generation reallocation implied by each counterfactual. Total generation can differ slightly across scenarios because the code permits non-zero shortage and oversupply in the renewable-counterfactual cases. In the regenerated outputs, S3 and S4 achieve exact energy balance with zero shortage and zero oversupply. S5 achieves zero shortage but introduces 75.9 GWh of oversupply due to min-load constraints under the endogenous commitment. All quantities are expressed in TWh and rounded to three decimals.

Table 18: Counterfactual Simulation: Revenue and Cost Decomposition

Scenario / Comparison	Thermal Revenue (bn RMB)	Fuel Cost (bn RMB)	Startup Cost (bn RMB)	Thermal Total Cost (bn RMB)	Wind Revenue (bn RMB)	Solar Revenue (bn RMB)
Panel A: Scenario levels						
1. Baseline (Observed)	32.584	37.270	0.089	37.359	4.620	1.892
2. S1 (Optimal Dispatch)	43.597	33.590	0.089	33.679	6.322	2.516
3. S2 (Optimal Commitment & Dispatch)	42.991	32.587	0.048	32.635	6.203	2.480
4. S3 (Fixed Commitment, Reference Dispatch)	30.005	33.144	0.089	33.233	6.379	2.712
5. S4 (Fixed Commitment, Economic Redispatch)	40.205	31.684	0.089	31.773	8.276	3.347
6. S5 (Endogenous Commitment & Redispatch)	40.054	30.726	0.039	30.765	8.240	3.346
Panel B: Selected cross-scenario changes						
Baseline → S1	11.013	-3.680	0.000	-3.680	1.702	0.624
S1 → S2	-0.606	-1.003	-0.041	-1.044	-0.119	-0.036
Baseline → S3	-2.579	-4.126	0.000	-4.126	1.759	0.820
S1 → S4	-3.392	-1.906	0.000	-1.906	1.954	0.831
S3 → S4	10.200	-1.460	0.000	-1.460	1.897	0.635
S4 → S5	-0.151	-0.958	-0.050	-1.008	-0.036	-0.001
S2 → S5	-2.937	-1.861	-0.009	-1.870	2.037	0.866
Baseline → S5	7.470	-6.544	-0.050	-6.594	3.620	1.454

Notes: Panel A reports scenario levels. Panel B reports changes relative to the comparison named in the first column. Thermal total cost equals fuel cost plus startup cost. All monetary values are expressed in billions of RMB and rounded to three decimals.

efficient allocation of thermal output across units and, in S2, from a more efficient commitment pattern. By contrast, the renewable-integration scenarios generate a large compositional shift. Relative to the baseline, S3 and S4 reduce thermal generation by 7.767 TWh while increasing wind generation by 5.297 TWh and solar generation by 2.470 TWh. The near one-for-one substitution from thermal to renewable output is the central physical margin through which the reform reshapes the generation mix. S5 preserves this broad reallocation, although endogenous recommitment increases thermal generation modestly relative to S4 and introduces some oversupply.

Table 18 then shows how these operational changes map into thermal-side revenue and cost. Under realized operations, thermal revenue is 32.584 billion RMB, fuel cost is 37.270 billion RMB, startup cost is 0.089 billion RMB, and thermal total cost is 37.359 billion RMB. Re-optimizing dispatch while holding the observed commitment set fixed (S1) lowers thermal total cost by 3.680 billion RMB, almost entirely through lower fuel use. Allowing commitment to adjust as well (S2) delivers an additional 1.044 billion RMB reduction in thermal total cost, with a concurrent reduction in startup cost of 0.041 billion RMB. Relative to the baseline, the combined gain from optimized dispatch and commitment is therefore 4.724 billion RMB, or 12.6% of baseline thermal cost. The decomposition thus indicates that the dominant engineering-cost saving operates through the intensive margin of dispatch, while the extensive margin contributes a smaller but still economically meaningful increment.

Table 16 shows that these engineering savings do not translate mechanically into lower user payments. Thermal accounting profit rises from -4.775 billion RMB in the baseline to 9.918 billion RMB in S1 and 10.356 billion RMB in S2, while the effective clearing price rises from 332.0 RMB/MWh to 459.7 RMB/MWh in S1 before easing to 452.5 RMB/MWh in S2. Consumer payment therefore rises substantially in S1 and remains well above the baseline in S2.

The final column of Table 16 reports total social surplus, defined as the sum of thermal and renewable accounting profits minus consumer payment—the standard welfare measure under perfectly inelastic demand, where $\Delta SS = \Delta \Pi_{th} + \Delta \Pi_{re} - \Delta \text{Payment}$. Every counterfactual transition yields a positive total surplus gain. Moving from the baseline to S1 (merit-order dispatch) raises total social surplus by 2.164 billion RMB, reflecting the engineering-cost savings from reallocating output to lower-cost units. The additional commitment optimization in S2 contributes a further 1.052 billion RMB. The renewable-integration counterfactuals are similarly welfare-improving: the baseline-to-S3 transition delivers a 2.060 billion RMB gain purely from displacing thermal output with zero-cost renewable generation, while the full reform bundle (Baseline to S5) raises total social surplus by 4.538 billion RMB. Crucially, the welfare gain from S5 is positive even though consumer payment rises by 12.534 billion RMB, because producer profits rise by more (thermal profit +14.064 billion RMB; renewable

profit +3.008 billion RMB). This decomposition underscores that rising consumer payment and rising social surplus are not mutually exclusive under the reform: the administrative benchmark price that prevailed before the reform was set below the efficient shadow cost of generation, so the transition to market pricing simultaneously raises both producer surplus and consumer expenditure while generating a net welfare gain through reduced engineering cost and expanded renewable accommodation.

Two institutional features of the pre-reform electricity market account for these patterns. First, the negative baseline thermal accounting profit (−4.775 billion RMB) reflects the system of administratively set benchmark prices in place before the spot market reform. The benchmark price of 332 RMB/MWh was set below the average total cost of thermal generation, including fuel costs and fixed capital charges, so that thermal generators operated at a book loss on the spot margin. Generators remained economically viable through medium-to-long-term forward contracts and regulated capacity payments, but the spot-settled revenue, which is the revenue concept recorded in the model, fell short of full cost recovery under the benchmark regime. Second, pre-reform policy mandated unconditional priority dispatch for renewable output, requiring thermal generators to accommodate renewable generation regardless of dispatch cost. This obligation compressed thermal generation hours and further reduced thermal spot revenue relative to potential. Together, these two forces—an administratively depressed clearing price and mandatory renewable accommodation—explain why baseline thermal accounting profit is negative in the structural model. The large jump in thermal profit and clearing price when switching to the merit-order scenarios (S1, S4, S5) reflects the replacement of the administratively fixed benchmark price with an endogenous market-clearing shadow price that equals the marginal cost of the last dispatched unit, which the structural model places substantially above the benchmark level. The Scenario Group I results should accordingly be read as engineering-cost gains in production and commitment rather than as a complete welfare ranking.

8.8.2 Renewable Integration and the Value of Flexibility

The renewable-counterfactual results are most naturally read in the same sequence. Table 17 first shows that the reform-induced renewable expansion primarily operates by displacing thermal generation. Relative to the baseline, both S3 and S4 increase renewable output by 7.767 TWh and reduce thermal output by 7.767 TWh. The near one-for-one substitution reflects the exact energy balance achieved by S3 and S4 in the regenerated outputs. In S5, renewable generation remains essentially unchanged relative to S4, but endogenous recommitment raises thermal output by 0.076 TWh and produces 75.9 GWh of oversupply due to min-load constraints.

Table 18 shows that this greener output mix does not imply a monotone reduction

in thermal-side cost under every institutional arrangement. Relative to the baseline, S3 lowers thermal total cost by 4.126 billion RMB because renewable output displaces thermal fuel consumption even without economic redispatch. The comparison between S3 and S4 isolates the value of within-commitment flexibility under a common renewable path and a common online fleet: thermal total cost falls by a further 1.460 billion RMB, entirely through lower fuel use, while startup cost is unchanged. The comparison between S4 and S5 isolates the additional value of recommitment. Thermal total cost falls by another 1.008 billion RMB, while startup cost also falls by 0.050 billion RMB, indicating that the endogenous commitment optimization reduces both fuel and cycling costs.

Table 18 reports the renewable-side counterpart. Relative to the baseline, moving to the counterfactual renewable path raises wind revenue by 1.759 billion RMB and solar revenue by 0.820 billion RMB at the baseline clearing price (S3), for a combined gain of 2.579 billion RMB. Under the endogenously clearing spot price in S4, the higher clearing price raises wind and solar revenues further: S4 wind revenue is 8.276 billion RMB and solar revenue is 3.347 billion RMB, compared with 6.322 and 2.516 billion RMB in S1. The combined renewable revenue increase from S3 to S4 is 2.532 billion RMB. The renewable revenue table captures the settlement-side gain from the greener generation mix and higher spot prices, separate from the thermal cost decomposition.

The final step is to examine incidence. Table 16 reports thermal profit, renewable profit, consumer payment, and the effective clearing price under each scenario. These are accounting objects within the model's settlement system, not welfare measures in the strict surplus-based sense. The comparison between *Baseline* and S3 isolates the mechanical effect of the renewable expansion without economic redispatch: thermal profit rises by 1.546 billion RMB, renewable profit rises by 0.514 billion RMB, and consumer payment is essentially unchanged because the effective clearing price remains at 332 RMB/MWh. The comparison between S1 and S4 is more economically revealing because both scenarios impose the same commitment set and differ only in the renewable path. Moving from S1 to S4 raises renewable revenue by 2.785 billion RMB. Thermal profit falls by 1.485 billion RMB, as the large decline in thermal generation (7.767 TWh) is accompanied by a decline in the effective clearing price from 459.7 to 444.9 RMB/MWh; consumer payment falls by 2.123 billion RMB. The downward pressure on the clearing price in S4 reflects the merit-order logic: renewable output displaces thermal generation, reducing the quantity dispatched from each thermal unit, which lowers each unit's marginal cost along the upward-sloping cost curve and therefore lowers the system shadow price lambda at equilibrium.

The same point appears in the comparison between S2 and S5, which is the most demanding test of cleaner renewable integration relative to the fully optimized observed-load benchmark. On the physical margin, the system is clearly greener in S5 than in

S2. On the renewable settlement margin, total renewable revenue is 2.903 billion RMB higher. Thermal total cost falls by 1.870 billion RMB in S5 relative to S2, as commitment optimization decommits the most expensive units in response to lower net thermal load, concentrating production among cheaper units and lowering average fuel cost. The reduction in the committed fleet's average cost also lowers the effective clearing price: the shadow price of the marginal committed unit falls from 452.5 RMB/MWh in S2 to 442.8 RMB/MWh in S5. Accordingly, the incidence table shows that thermal profit is 1.067 billion RMB lower, consumer payment is 1.552 billion RMB lower, and the effective clearing price is 9.686 RMB/MWh lower in S5 than in S2. The renewable expansion therefore delivers a greener generation mix together with lower engineering cost and lower consumer payments relative to the fully optimized observed-load benchmark, as commitment optimization concentrates thermal production among the cheapest available units.

The *Baseline-to-S5* comparison summarizes the full-system transformation implied by the reform. Relative to observed operations, S5 reduces thermal generation by 7.691 TWh and raises renewable generation by 7.768 TWh, with wind accounting for 5.298 TWh and solar for 2.469 TWh. On the thermal side, revenue rises by 7.470 billion RMB while thermal total cost falls by 6.594 billion RMB, because fuel savings of 6.544 billion RMB are accompanied by a startup cost reduction of 0.050 billion RMB. On the renewable side, aggregate renewable revenue rises by 5.074 billion RMB. In market-accounting terms, thermal profit increases by 14.064 billion RMB, renewable profit increases by 3.008 billion RMB, consumer payment increases by 12.534 billion RMB, and the effective clearing price rises by 110.8 RMB/MWh. This final comparison demonstrates that the reform bundle combines a large reallocation toward cleaner generation and lower thermal engineering cost with higher consumer payments and a higher effective clearing price, so that the incidence of the welfare gain is uneven: consumer payment rises while producer accounting profits rise by more, and the engineering-cost savings and their distributional consequences must therefore be assessed separately.

Finally, one caveat regarding the structural simulation framework warrants acknowledgment. The counterfactual models abstract from exogenous physical contingencies such as planned maintenance, forced outages, and detailed transmission network constraints. The optimized scenarios should therefore be interpreted as disciplined engineering-accounting counterfactuals rather than literal operating forecasts. That qualification does not alter the central message of the results: merit-order dispatch and commitment optimization generate large engineering-cost savings, while the cost consequences of renewable integration depend critically on operational flexibility.

9 Conclusion and Policy Implications

This paper examines the full-scale electricity spot market reform in a northern Chinese province and finds that high-frequency price signals rapidly reshaped dispatch incentives, reduced daily thermal generation by 12.1%, and raised wind and solar accommodation by 38.5% and 43.8%, respectively. Combining a Regression Discontinuity in Differences (RD-DD) design with a dynamic structural framework, we show that these aggregate outcomes reflect heterogeneous unit-level adjustment and are accompanied by substantial engineering-cost savings, but also by a simultaneous rise in consumer payment under market-based pricing rather than by an unambiguous improvement in every welfare dimension.

Our reduced-form estimates show that the aggregate reallocation is sharply heterogeneous: spot pricing displaced generation most forcefully from older, larger, and higher-cost thermal units, while smaller and more flexible units increased their conditional output by shifting into peaking roles. The extensive margin—the probability of being dispatched—accounts for the bulk of the aggregate thermal reduction, while the intensive margin reflects the increased output concentration among units that remain dispatched.

To interpret these patterns, we recover unit-level marginal costs and startup costs for the thermal fleet within a dynamic structural model. The institutional setting and model-comparison results are consistent with spot-market outcomes that are approximately competitive under extensive forward contracting. Embedding the recovered cost structure in the counterfactual simulations yields three main conclusions. First, optimized dispatch and commitment generate engineering-cost savings relative to observed operations, with the intensive margin of dispatch accounting for the larger share. Second, the reform-induced renewable expansion produces a markedly cleaner generation mix and a large reduction in thermal generation; relative to the fully optimized observed-load benchmark, commitment optimization concentrates thermal production among cheaper units under lower net load, delivering an additional reduction in engineering cost. Third, accounting profits, consumer payment, and effective clearing prices do not move proportionally with engineering cost. For that reason, we interpret thermal and renewable profits, consumer payment, and clearing prices as market-accounting outcomes rather than as a complete welfare ranking.

These findings have several policy implications for electricity-market liberalization, especially in developing and transition economies. First, the counterfactual exercise highlights a central operational trade-off: the value of additional renewable generation depends on the system’s ability to adjust thermal operations without creating excessive startup costs, minimum-load inefficiencies, or balancing frictions. Energy-only spot markets strengthen incentives for flexibility, but may not fully compensate thermal units

for the discrete startup costs and long-run capital degradation associated with increased cycling under renewable integration. This suggests a role for complementary flexibility mechanisms, including ancillary-service markets and, where appropriate, capacity remuneration arrangements. The Baseline-to-S5 comparison underscores this point: the reform bundle delivers a cleaner generation mix and lower thermal engineering cost, yet consumer payment rises in the model's settlement accounting because the endogenous clearing price substantially exceeds the pre-reform administered benchmark. More broadly, the results illustrate a general principle for policy evaluation: engineering-cost savings, market-accounting incidence, and social welfare in the strict surplus-based sense are conceptually distinct and must be assessed separately, rather than collapsed into a single metric. The reform studied here shows that all three can move in different directions simultaneously.

Second, this provincial experience suggests that the *dual-track* market design can limit spot-market power incentives during the transition to liberalized dispatch. Extensive forward contract coverage reduced the scope for strategic withholding relative to settings studied in some early deregulated markets. More generally, the evidence is consistent with the view that transitional market institutions—such as the dual-track CfD architecture studied here—can support cleaner dispatch and lower engineering cost even before a fully mature wholesale market architecture is in place, although a formal comparison to more advanced designs is beyond the scope of this paper.

Finally, our study has limitations that point to several directions for future research. The structural counterfactuals abstract from some localized physical contingencies, including detailed transmission network constraints, planned maintenance, and forced outages. Future work could integrate richer network models, such as Optimal Power Flow (OPF) frameworks, into a dynamic strategic environment. Even with these limitations, the combined reduced-form and structural evidence indicates that high-frequency market mechanisms can reduce thermal engineering cost and accelerate renewable integration—a finding with direct policy relevance for developing economies that are scaling renewable capacity while relying on coal-dominated thermal fleets for system reliability.

A Structural Estimation: Robustness and Supplementary Tables

A.1 Elasticity Bandwidth Sensitivity

Table 19 reports the median unit-level residual demand elasticity from the Nadaraya-Watson kernel estimator (Method A–Unit) for five bandwidth values ranging from $h = 30$ to $h = 400$ RMB/MWh. The baseline choice of $h = 80$ is selected by cross-validation. The median elasticity is relatively stable between $h = 50$ and $h = 80$, while very small bandwidths ($h = 30$) introduce high variance and very large bandwidths ($h = 150$, $h = 400$) over-smooth the local slope, attenuating the estimated elasticity toward zero. The baseline choice therefore balances bias and variance; the directional conclusion (significantly negative residual demand elasticity) is robust across all bandwidths shown.

Table 19: Elasticity Estimation: Bandwidth Sensitivity

Bandwidth (h)	30	50	80	150	400
Median elasticity (MWh / [RMB/MWh])	−40.05	−33.05	−36.35	−14.34	−5.35
N observations	9,615,360				

Notes: All bandwidths use the same unit-level Nadaraya-Watson kernel estimator (Method A–Unit). Very small bandwidths ($h = 30$) yield highly variable estimates due to sparse local support; very large bandwidths attenuate the slope toward zero. The primary specification uses $h = 80$.

A.2 Startup Cost Estimation: Horizon Sensitivity

Table 20 compares the distribution of unit-level startup cost estimates across three forward simulation horizons: $H = 96$ (24 hours), $H = 192$ (48 hours, primary), and $H = 384$ (96 hours). The $H = 192$ and $H = 384$ distributions are virtually identical in all summary statistics, while $H = 96$ yields somewhat higher median and mean estimates (RMB 82,400 vs. RMB 65,800) and a heavier right tail. This suggests that the shorter horizon over-weights near-term cycling opportunities, leading to higher imputed option values and therefore higher recovered startup costs. The $H = 192$ horizon is preferred as it balances computational stability against long-run mean-reversion.

A.3 Structural Estimates of Startup Costs: Pooled Branch

Table 21 reports the structural parameters recovered via the SMD estimator for the 29 Branch B units whose on/off transition counts were insufficient for individual identification ($N_{trans} < 30$). Moments are pooled within three capacity tiers. The

Table 20: Startup Cost Estimates: Sensitivity to Simulation Horizon

Horizon	N	Mean (RMB 10k)	Median (RMB 10k)	Std. Dev. (RMB 10k)	P10 (RMB 10k)	Max (RMB 10k)
Panel A: Full fleet						
$H = 96$ (24 hrs)	114	26.31	8.24	42.92	2.51	274.66
$H = 192$ (48 hrs, primary)	114	16.59	6.58	21.14	2.48	111.76
$H = 384$ (96 hrs)	114	16.61	6.58	21.17	2.49	112.07
Panel B: Small units (< 300 MW)						
$H = 96$	31	14.17	5.23	–	–	77.20
$H = 192$	31	9.32	4.39	–	–	40.15
$H = 384$	31	9.36	4.40	–	–	40.22
Panel C: Medium units (300–600 MW)						
$H = 96$	58	24.34	7.24	–	–	163.57
$H = 192$	58	14.00	6.27	–	–	65.75
$H = 384$	58	13.98	6.27	–	–	65.29
Panel D: Large units (≥ 600 MW)						
$H = 96$	25	45.95	18.48	–	–	274.66
$H = 192$	25	31.62	17.03	–	–	111.76
$H = 384$	25	31.68	17.04	–	–	112.07

Notes: All estimates use the B-spline Probit Hybrid Algorithm with $N = 500$ simulation paths. Monetary values are expressed in units of RMB 10,000. All three horizons are evaluated on the same balanced sample of 114 units; one Small unit (< 300 MW, 50 MW installed capacity) that did not converge at $H = 192$ is excluded from all horizons to ensure comparability. $H = 192$ and $H = 384$ produce nearly identical distributions across all groups, confirming robustness. “–” indicates within-group standard deviations and percentiles are available upon request.

estimates reveal a pronounced capacity gradient: for Small and Medium units the startup cost rises with installed capacity ($\beta_{cap} > 0$), while for Large units the positive intercept dominates and the capacity coefficient turns negative, reflecting that the per-MW startup cost increment diminishes at the high end of the capacity range. The Medium-group base constant is numerically negligible ($\approx 3.7 \times 10^{-18}$), indicating that startup costs for that tier are driven almost entirely by the capacity and age channels. The Age Effect (β_{age}) is negative for Small and Large units, suggesting that startup costs decline with unit age within these size groups—potentially reflecting that units surviving to older age in these capacity tiers tend toward simpler boiler configurations. The positive age coefficient for Medium units is more consistent with thermal degradation increasing cycling costs.

Table 21: Structural Estimates of Startup Costs: Pooled Branch ($H = 192$ Horizon)

Capacity Group	Base Cost Constant (β_0)	Capacity Effect (β_{cap} , per 100 MW)	Age Effect (β_{age} , per 10 Years)	Shock Variance (σ)
Small Units (< 300 MW)	67,429.2	3,211.1	-14,579.4	17,009.4
Standard Error	(4,539.9)	(1,878.7)	(1,586.5)	(690.0)
Medium Units (300–600 MW)	$\approx 0^\dagger$	15,871.4	5,377.2	20,746.4
Standard Error	(21,529.3)	(5,831.1)	(3,372.9)	(821.8)
Large Units (≥ 600 MW)[‡]	1,283,602.0	-161,379.9	-90,629.9	47,471.0
Standard Error	(94,958.2)	(14,215.2)	(13,049.1)	(2,553.4)

Notes: This table reports the structural parameters recovered via the Simulated Minimum Distance (SMD) estimator (Branch B of the two-branch estimation strategy) using the $H = 192$ quarter-hour forward simulation horizon. The dependent variable is the latent startup cost measured in RMB. Standard errors are derived from the numerical Hessian of the WNLLS objective function. [†] The Large-unit group (≥ 600 MW) contains 25 units. [‡] The Medium-group base constant is numerically negligible ($\approx 3.7 \times 10^{-18}$), indicating that the intercept is not separately identified from the capacity and age effects for this tier; startup costs for Medium units are therefore driven almost entirely by the capacity (β_{cap}) and age (β_{age}) channels. Sensitivity results for $H = 96$ and $H = 384$ horizons are reported in Appendix Table 20.

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